

claude-windows / c1fcb4aa-537d-41fd-858e-53f93349bf5a

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\subagents\workflows\wf_41c8dc8a-8a4\agent-a255088bcb796f154.jsonl`  
- SHA-256: `20edc9f8220766aa23db9374ecd05f27f1e15f6682bde2968a7c2bece39ebc46`  
- Source modified: `2026-07-02T02:53:47+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_41c8dc8a-8a4`  
- session_id: `c1fcb4aa-537d-41fd-858e-53f93349bf5a`
```

Transcript

1. user (2026-07-02T02:47:11.977Z)

CONTEXT (read carefully):

- You are one auditor in a multi-agent tech-debt/critical-fix audit of the khelsutra-guru multi-repo workspace at C:/Users/avidu/Projects/khelsutra-guru. Today is 2026-07-02. All git repos were just pulled to current origin master/main.
- Repos: badminton-highlight-indexer (the ENGINE, Python, ~1500 tests, org Khelsutra), khelsutra (SPA web/ + marketing site/, TS/React, org avidu), sports-data-collector, rally-annotator, sports-obsreport (small Python repos). Plus two NON-git dirs: "Annotation Setup", khelsutra-screencast.
- HARD RULES: READ-ONLY. Do NOT modify/create/delete any file, do NOT create branches/worktrees, do NOT pip install, do NOT run the full test suite (CI is trusted on master). You MAY run read-only git and gh commands (gh is authenticated), ruff/mypy IF already installed, and fast read-only scripts.
- EVERY claim must cite file:line (or a gh URL). A claim without file:line evidence is a hypothesis - mark it as such or drop it. The codebase drifts between sessions: re-verify anything you believe from docs against actual code.
- OWNER-GATED areas (flag as owner_gated=true, still report): alpha/serving go-live + Cloudflare dashboard work, Studio P10/P11 (corpus promotion + train/reeval), product/strategy/licensing decisions, real GPU/cloud spend, counsel ToS/DPA, repo rename off "badminton".
- Known-open items from project memory (VERIFY current state, don't trust blindly): engine PR #390 (D1 split main.py, open since 06-26) · audit doc docs/CODE_CLEANUP_AUDIT_2026-06-24.md remaining items D1, D13 (ByteTrack->trackers), Phase-3 (B3/B4/B6/B7/B8/D5/D6/D8-D10/O3/O4) · docs/DECISION_SNAPSHOT_REGRESSION.md next=P1 · docs/GOLDEN_REGRESSION_FIXTURES.md backlog M4/P0-rename/O1/O2 · engine issues #340 (Gemini re-bill), #332 (NSFW preflight), #349 (CPU-bound decode), #384 (CI single-source) · khelsutra issues #97-#114 UX/alpha backlog · DO droplet RETIRED 2026-06-25 (docs may still reference it; api.khelsutra.guru CNAME/CF-Access cleanup was pending).
- Categorize each finding: critical-fix (latent bug/correctness/data-loss/cost-leak), security, tech-debt (structure/duplication/dead code), test-gap, ci-infra, doc-rot, hygiene.
- Severity: P0 = latent bug or cost/data/security risk reachable in normal use; P1 = high-value debt blocking velocity or a stale-open thread; P2 = worthwhile cleanup; P3 = nice-to-have (report max 3 of these).
- Be selective: max ~12 findings, each genuinely worth acting on. Note tracked_in when an issue/PR/doc already tracks it (tracked is NOT done - still report). Your final message is data for an orchestrator, not prose for a human.

YOUR DIMENSION: reconcile the ENGINE's own tracking docs against current code. Repo: C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer. Read docs/CODE_CLEANUP_AUDIT_2026-06-24.md, docs/BACKLOG.md, docs/NEXT_STEPS.md, docs/DOC_STATUS.md, docs/DECISION_SNAPSHOT_REGRESSION.md (section 7), docs/GOLDEN_REGRESSION_FIXTURES.md (section 9). For every item those docs claim is OPEN/remaining, verify against current code at file:line whether it is (a) still open, (b) silently done (doc-rot), or (c) obsolete. Report the highest-value still-open items as findings and any doc-vs-reality drift as doc-rot findings. Also check: does open PR #390 (gh pr view 390)

still apply cleanly / is it stale vs master?

2. user (2026-07-02T02:47:18.747Z)

e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
81a78bf api: /api/videos returns video_id + a friendly match_name (fix disappearing name) (#412)
8b16cf4 docs: reflect today's storage endpoints + the corpus/user-video boundary (#411)

3. user (2026-07-02T02:47:19.569Z)

docs\ABLATION_LAYER.md
docs\ANALYZERS_AND_RECONCILERS.md
docs\CODE_AUDIT_AND_TEST_HARDENING.md
docs\DEFERRED_HARDENING_PLAN.md
docs\EVALUATION.md
docs\EXECUTION_POLICY.md
docs\HARDENING_LOOP_HANDOFF.md
docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md
docs\PERSONALIZATION_PLAN.md
docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md
docs\STORAGE_SHARING_MODEL.md
docs\TRACKNET_WSL_SETUP.md
docs\UI_ALPHA_EXECUTION_PLAN.md
docs\VIDEO_LOCALITY_MODEL.md
docs\BACKLOG.md
docs\CLOUD_GPU_DRYRUN_GUIDE.md
docs\COMMERCIALIZATION.md
docs\CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md
docs\CREATING_REELS.md
docs\DESKTOP_APP_PLAN.md
docs\EXPERIMENT_HARNESS.md
docs\GOLDEN_REGRESSION_FIXTURES.md
docs\HOSTING_PLAN.md
docs\HOW_RALLY_DETECTION_WORKS.md
docs\MULTI_SIGNAL_FUSION_PLAN.md
docs\PACKAGING_PLAN.md
docs\PER_VIDEO_WORKSPACE.md
docs\PLATFORM_ARCHITECTURE.md
docs\PROJECT_DOC_TEMPLATE.md
docs\PROMINENT_COURT_DETECTION.md
docs\R1_MILESTONE_LAUNCH_REPORT.md
docs\R2_EVAL_METRIC_DESIGN.md
docs\RALLY_DETECTION_GAP_CLOSURE.md
docs\RALLY_DETECTION_QUALITY_REPORT.md
docs\RALLY_QUALITY_RESEARCH.md
docs\REAL_FOOTAGE_VALIDATION_REPORT.md
docs\ROADMAP.md
docs\SETUP_NEW_MACHINE.md
docs\UI_PROPOSAL.md
docs\KHELSUTRA_PRODUCT_OVERVIEW.md
docs\DECISION_SNAPSHOT_REGRESSION.md
docs\README.md
docs\CODE_CLEANUP_AUDIT_2026-06-24.md
docs\NIGHTLY_ENGINE_REGRESSION.md
docs\RALLY_DETECTION_CODE_REVIEW.md
docs\CODE_MAP.md
docs\DATA_IN_GCS.md
docs\DOC_STATUS.md
docs\I18N_PLAN.md
docs\NEXT_STEPS.md
docs\QUALITY_ITERATIONS.md
docs\RALLY_DETECTION_FIXES_PLAN.md

4. user (2026-07-02T02:47:27.707Z)

```

1      # Documentation Status & Archival Register
2
3      > **Status:** ? **LIVING** ? the single board for doc **lifecycle + archival due-
diligence**. **Last full review:** 2026-06-21.
4      > **Update cadence:** whenever a doc changes lifecycle state (created / shipped a
milestone / delivered / archived), and re-review at each clean-slate.
5      > **Why this exists:** so we can (a) **review completeness** at a glance, (b) **update
it as work finishes** to prove we're done, and (c) **never archive a foundational doc without
honestly updating it first** (owner directive, 2026-06-21).
6
7      ---
8
9      ## 0. How to use this
10     - **Reviewing now?** Read §4 (the open findings/actions) ? that's what needs attention.
§3 is the full register.
11     - **Marking something complete?** Update the doc's own **§7 progress tracker** (the
source of truth for *its* completeness), then flip its row here, then ? if terminal ? run the §2
archival checklist.
12     - **This register does NOT duplicate completeness detail** ? it indexes lifecycle +
archive-readiness and points to each tracked doc's §7.
13
14     ## 1. Lifecycle legend
15     `DRAFT ? IN PROGRESS ? DONE/DELIVERED ? ARCHIVED`. Plus: **LIVING** (permanent
reference, never "done"), **PROPOSAL/DESIGN** (owner-gated, not started), **REPORT** (a terminal
record of a run/launch), **CONVENTION** (a standing rule/guide).
16
17     ## 2. Archival due-diligence checklist ? *the repeatable process (do NOT just `git
mv`)*
18     Before a foundational/tracked doc moves to `docs/archives/`:
19     1. **VERIFY claims against the code** ? no stale assertions; tag load-bearing facts
`[verified]`. (A doc that says "nothing built" when code shipped is *not* archive-ready ? fix it
first.)
20     2. **Flip Status ? DELIVERED / DONE / REJECTED + a completion note** ? what shipped,
where (PRs, paths, artifacts), and what was **deferred / carried forward**.
21     3. **Update ALL inbound references** ? `docs/README.md` index, `CLAUDE.md`,
`docs/NEXT_STEPS.md`, and any cross-doc links ? the new path + status.
22     4. **Keep the lineage** ? ADR forward/back-links + "supersedes / superseded-by" so the
evolution stays traceable.
23     5. **`git mv` the whole dir/file ? `docs/archives/past_projects/`** (terminal-state
ONLY; live docs stay at `docs/` top level). Relative links inside become historical (note this
in the doc's header).
24     6. **Record the move** ? flip the row here to §3e + a line in memory `current-state`.
25
26     > **Worked example (the template):** **DECOUPLED_COMPUTE ? #270** ? verified M0?M2+M3.5
delivered on GCP, flipped ??? DELIVERED with a completion note (incl. carried-forward
WASB-C3/DPA/`md5`), reindexed `docs/README` + `past_projects/README`, kept the ADR-010?014
lineage, then `git mv` to `docs/archives/past_projects/DECOUPLED_COMPUTE/`.
27
28     ## 3. The register
29
30     ### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's §7 is the
completeness source-of-truth)
31     | Doc | Status | Archive-ready? | Next action |
32     |---|---|---|---|
33     | `COMPUTE_DECOUPLED_SERVING/` | **IN PROGRESS** ? M0 approved + **M1?M4 SHIPPED**
(#279/#280/#282/#283); M5+ default-OFF code batch authorized (2026-06-21) | No | M5 truly-local
? M6 Cloud Run ? M8 ledger ? M9-auth ? M11 flywheel ? M12 gdrive-api (owner residual:
counsel/prices/OAuth/real-run within the D17 cap) |
34     | `RALLY_QUALITY_RESEARCH.md` | Foundation, **tracked & ACTIVE** (§7) | No | Drives the
R-series; complete when §7.1 single quality number is attributed |
35     | `R2_EVAL_METRIC_DESIGN.md` | IMPLEMENTED (augmenting) | No (gates R4 + dual-eval;
ablation pending) | Promote-or-reject toLf1 as the gate (ablation) ? then archive with the
R-cluster |
36     | `R1_MILESTONE_LAUNCH_REPORT.md` | LAUNCHED (#204) ? terminal record | Soon (with the
R-cluster) | Keep with R-series for narrative; archive when the track wraps |

```

37 | `RALLY\_DETECTION\_GAP\_CLOSURE.md` | DRAFT (owner-gated) | No | G1?G3 levers; part of the R/quality cluster |

38 | `RALLY\_DETECTION\_FIXES\_PLAN.md` | ****DONE**** ? findings A/B/C resolved; P2b promoted `inactivity\_temporal\_density` default-ON (#396) | With the R-cluster (deferred per its header; gate = R2 toI1F1 ablation + \$7.1 number) | Archive with the R-series when the track wraps |

39 | `RALLY\_DETECTION\_CODE\_REVIEW.md` | Terminal snapshot ? findings A (#394), B (#394 ? promoted #396), C (`sdc#50`) all resolved | With the R-cluster | Archive with the R-series |

40 | `PER\_VIDEO\_WORKSPACE.md` | DRAFT, IN PROGRESS (P0 = #264) | No | ? supersede-reconcile (finding #1) DONE; P3 de-hardcode base ? via #282; remaining P1?P6 (P1 v6 migration coordinates with M8's v6 cost-ledger ? version bump authored once) |

41 | `MULTI\_SIGNAL\_FUSION\_PLAN.md` | P1 DONE; P2+ owner-gated | No | P3/P4 (learned fusion, audio) ? needs GPU golden features |

42 | `EXPERIMENT\_HARNESS.md` | P1 IMPLEMENTED | No | A/B + promotion gate; supports the quality track |

43 | `PROMINENT\_COURT\_DETECTION.md` | DRAFT (owner-gated) | No | C-series; multi-track selection design |

44 | `EXECUTION\_POLICY.md` | P1 SHIPPED; P3b next | No | Async job policy; reused by Cloud Run dispatch |

45 | `GOLDEN\_REGRESSION\_FIXTURES.md` | IN PROGRESS (owner-gated) | No | Golden eval fixtures |

46 | ****Data layer**** ? `docs/data\_pipeline/` (`GOLDEN\_SET\_IMPLEMENTATION` . `GOLDEN\_SET\_PHASE1\_VIDEOS` . `GOLDEN\_VIDEOS` . `VIDEO\_RECORDING\_GUIDELINES` . `GOLDEN\_DATA\_SHARING`) | LIVING (data, ****no-ML****) | n/a | Relocated 2026-06-21 to the isolated data pillar; tracked via `data\_pipeline/README.md`. Roora **vendor** docs archived ? `archives/roora-vendor/`. |

47 | `CODE\_CLEANUP\_AUDIT\_2026-06-24.md` | ****IN PROGRESS**** ? Phase 1: P0/P1/P4/P6a shipped (#352/#355/#354/#356). Quick wins (Q4/Q5/Q6/Q8) next. | No | Q4?Q5?Q6?Q8 ? P6b ? P2 (segmenters) ? P3 (extra=forbid) ? P5 (app-factory) ? P7 verify |

48 | `DESKTOP\_APP\_PLAN.md` | DRAFT (owner-gated, nothing built) | No | P0 compute-feasibility spike; relates to truly-local profile |

49 | `OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` | PLAN/roadmap (owner-gated) | No (authoritative roadmap) | Gen-0?Gen-2; the owned-model track |

50 | `PERSONALIZATION\_PLAN.md` | DESIGN ? owner-adopted; Phase-0 | No | Per-user models (DB v5 store landed) |

51 | `UI\_ALPHA\_EXECUTION\_PLAN.md` | ACTIVE | No | khelsutra UI alpha track |

52 | `I18N\_PLAN.md` | Design (P0); ****in implementation**** (adapted) | No | Executed in `khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md` (P1?site/, P3?web/); P2 backend stays here, owner-gated |

53 | `POST\_166...RISK\_ASSESSMENT.md` | Plan for owner review | No | Milestones/risk doc |

54

55 **### 3b. Foundational / strategy references (LIVING ? archive only if ****superseded****, not when "done")**

56 | Doc | Status | Note |

57 | ---|---|---

58 | `PLATFORM\_ARCHITECTURE.md` | Strategy/research | The `ComputeBackend`/`StorageBackend` registries ? realized by COMPUTE_DECOUPLED_SERVING |

59 | `COMMERCIALIZATION.md` | ? LIVING | Strategy; carries WASB-C3 |

60 | `VIDEO\_LOCALITY\_MODEL.md` | THINKING DOC | L0 fully-local = the truly-local profile |

61 | `HOSTING\_PLAN.md` | PROPOSAL | ****?** partly overtaken by ADR-013 + the shipped site ? verify (finding #5)****** |

62 | `UI\_PROPOSAL.md` | PROPOSAL (partly shipped) | UX half ****in implementation**** ? `khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md`; currency banner added (finding #5 ? for this doc) |

63 | `STORAGE\_SHARING\_MODEL.md` | DRAFT (co-design pending) | ? |

64 | `ANALYZERS\_AND\_RECONCILERS.md` | DESIGN (owner-gated) | ? |

65 | `ROADMAP.md` / `CODE\_MAP.md` / `HOW\_RALLY\_DETECTION\_WORKS.md` / `KHELSUTRA\_PRODUCT\_OVERVIEW.md` | LIVING references | Keep current; not archive targets |

66

67 **### 3c. Reports & terminal records (archive candidates once their track wraps)**

68 | Doc | Status | Note |

69 | ---|---|---

70 | `REAL\_FOOTAGE\_VALIDATION\_REPORT.md` | validation complete | Motivates R2; archive with the R-cluster |

71 | `RALLY\_DETECTION\_QUALITY\_REPORT.md` | report | Quality snapshot; part of R-cluster |

72 | `QUALITY\_ITERATIONS.md` | LIVING empirical log | ****Keep**** ? the paper-backbone

ablation log ([[paper-coauthor-aim]]) |

73 | `CODE\_AUDIT\_AND\_TEST\_HARDENING.md` · `HARDENING\_LOOP\_HANDOFF.md` ·
 `DEFERRED\_HARDENING\_PLAN.md` | COMPLETED ? **top-level pointer stubs** (archived copies exist) |
 ? confirm intentional stubs (per #165) vs finish-archiving (finding #4) |

74

75 ### 3d. Conventions / guidelines / templates / runbooks (permanent ? do not archive)

76 `PROJECT\_DOC\_TEMPLATE.md` · `NEXT\_STEPS.md` (? refreshed 2026-06-21) · `BACKLOG.md` ·
 `EVALUATION.md` · `ABLATION\_LAYER.md` · `SETUP\_NEW\_MACHINE.md` · **`CREATING\_REELS.md`** (? end-
 user usage guide ? install ? process ? pick ? reel; LIGHT tier; added 2026-06-23) ·
 `CONVERT\_GCS\_VIDEOS\_TO\_HIGHLIGHTS.md` (? operator runbook ? `gs://` videos ? reels on L4
 Cloud Run; verified e2e 2026-06-23) · `TRACKNET\_WSL\_SETUP.md` · `README.md`. *(2026-06-21
 relocation: the data-layer guides moved ? `data\_pipeline/`; the field-ops + Roora docs ?
 `archives/{field-pmf,roora-vendor}/`.)*

77

78 ### 3e. Already archived (`docs/archives/`)

79 `past\_projects/DECOUPLED\_COMPUTE/` (? 2026-06-21, #270) · `past_projects/low-regret-
 moves/` · \*\*`field-pmf/`\*\* (field-ops + customer-feedback, 2026-06-21) · \*\*`roora-vendor/`**
 (annotation-vendor engagement, 2026-06-21) · `PER\_VIDEO\_OUTPUT\_LAYOUT-SUPERSEDED-2026-06-21.md`
 · `CODE\_AUDIT\_AND\_TEST\_HARDENING-COMPLETED-2026-06-14.md` · `DEFERRED_HARDENING_PLAN-
 COMPLETED-2026-06-14.md` · `HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` ·
 `COMMERCIAL\_READINESS\_REVIEW.md` · `MONETIZATION\_AUDIT.md` · `TEST\_RUN\_SUMMARY.md` · `research/`
 · `checkpoints/` · `quality-iterations/`.

80

81 ### 3f. ADR log ? `docs/archives/decisions/DECISIONS.md`

82 **ADR-001 ? ADR-014** (14, all ratified). The compute/serving lineage: **ADR-005 ? 008 ?
 009 ? 010 ? 011 ? 012 ? 013 ? 014** . ADR-014 was **RATIFIED 2026-06-21** (owner M0 sign-off for
 COMPUTE_DECOUPLED_SERVING).

83

84 ## 4. Open due-diligence findings / actions (2026-06-21 review)

85

86 > **? Doc-restructuring done (2026-06-21, owner-directed):** field-ops/customer-feedback
 ? `archives/field-pmf/`, Roora annotation-vendor ? `archives/roora-vendor/`, and a no-ML **data
 layer** carved into `docs/data\_pipeline/`. Inbound + outbound links rewired; three dir READMEs
 added (the `data\_pipeline` one sets the "no segmentation/stitching" boundary).

87 1. **? RESOLVED (2026-06-21) ? `PER\_VIDEO\_WORKSPACE.md` supersedes
 `PER\_VIDEO\_OUTPUT\_LAYOUT.md`.** Folded forward the `video\_slug` naming option (workspace \$8.5) +
 the `output/GX010125\_run/` manual-prototype precedent; predecessor archived ?
 `docs/archives/PER\_VIDEO\_OUTPUT\_LAYOUT-SUPERSEDED-2026-06-21.md`; `CODE\_MAP.md` repointed.

88 2. **? RESOLVED (2026-06-21) ? `NEXT\_STEPS.md` refreshed** with a current 2026-06-21
 state + a **cross-track prioritized pending list** ; the 2026-06-13 owned-model/quality body is
 kept (clearly marked) as that track's detail.

89 3. **R-series / quality cluster is ACTIVE** (`R1`, `R2`, `RALLY\_DETECTION\_GAP\_CLOSURE`, `RA
 LLY_QUALITY_RESEARCH`, `QUALITY_ITERATIONS`, `REAL_FOOTAGE_VALIDATION_REPORT`, `RALLY_DETECTION_QUA
 LITY_REPORT`). **Do NOT archive yet** ? archive as a *cluster* once R2 is promoted-or-rejected
 as the gate + \$7.1's single number is attributed. (Owner already confirmed "leave as is",
 2026-06-21.)

90 4. **Confirm the COMPLETED top-level stubs** (`CODE\_AUDIT\_AND\_TEST\_HARDENING`,
 `HARDENING\_LOOP\_HANDOFF`, `DEFERRED\_HARDENING\_PLAN`) are intentional pointer-stubs (per #165) ?
 or finish-archive them so there's one copy.

91 5. **`HOSTING\_PLAN.md` / `UI\_PROPOSAL.md` are PROPOSAL but partly overtaken** by ADR-013
 (DO dropped) + the shipped khelsutra site. **? `UI\_PROPOSAL.md` banner added (2026-06-21)** ?
 its UX half is now tracked in `khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md`; the `I18N\_PLAN.md`
 localization track is likewise in implementation there. **`HOSTING\_PLAN.md` banner still
 pending** (ADR-013 already noted in it).

92 6. **? RESOLVED (2026-06-21) ? `COMPUTE\_DECOUPLED\_SERVING` M0 approved + M1?M4 SHIPPED**
 (#279/#280/#282/#283); all \$8 gating Qs locked (D7?D17); M5+ default-OFF code batch authorized
 (owner residual: counsel ToS/DPA, real GCP prices, OAuth app, real cloud run within the D17
 \$100/?10k cap).

93

94 ## 5. Maintenance

95 Update a row when its doc changes lifecycle; run \$2 before any archive; re-review \$4 at
 each clean-slate. This register + the \$2 checklist are the operational form of the `CLAUDE.md`
 archive policy and [[working-agreement-tracked-project-docs]].

96

5. user (2026-07-02T02:47:28.004Z)

```
1      <!--
2      PROJECT DOC ? tracked code-cleanup + audit remediation plan.
3      Follows the PROJECT_DOC_TEMPLATE.md conventions.
4      -->
5
6      # Code Cleanup ? 2026-06-24 Deep Audit Remediation
7
8      > **Status:** `PHASE 1 COMPLETE` ? 14 items shipped + D2 app-factory + test fixes .
**Owner:** Avi . **Created:** 2026-06-24 . **Last updated:** 2026-06-25 (DoD/line-count accuracy
fixes ? see Changelog)
9      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to
`docs/archives/past_projects/` when DONE)
10     > **Tracking anchors:** $7 progress tracker is the source of truth; indexed in
`docs/README.md`.
11     > **Relation to existing docs:** Sibling of `docs/CODE_AUDIT_AND_TEST_HARDENING.md` (the
2026-06-10 hardening sweep).
12     >
13     > **Phase 1 summary:** All structural refactorings landed. `backend.api.job_worker`
lifted from mypy quarantine. Suite: **1463/1463 green**. `ruff`: all checks passed. `mypy`:
quarantined modules unchanged (ratchet held).
14
15     ---
16
17     ## 0. TL;DR
18
19     A deep, whole-repo audit on 2026-06-24 surfaced **35 items** (33 from fresh review + 2
absorbed from `docs/BACKLOG.md`) across bugs, technical debt, operational risks, and quick wins.
This doc organizes them into a phased cleanup project: **Phase 1 = pure refactorings** (no logic
change, all tests must stay green, coverage must not drop), **Phase 2+ = logic fixes +
hardening**. The first approved targets are **B2, D3, D2, B5** ? the `Success`/`Failure`
contract for Gemini, typed-config migration completion, app-factory refactor, and the no-op
segmenter validator fix.
20
21     ---
22
23     ## 1. Problem & goal
24
25     The codebase has grown rapidly since the June 2026 scaffolding and has accumulated
several patterns that make it harder to extend safely:
26
27     - **Silent failure masking** ? `analyze_video()` returns `[]` on ANY error (missing API
key, network failure, model crash) indistinguishable from "video has no rallies."
28     - **Dual config access** ? new code uses typed `AppConfig` attribute access; old code
(validators, segmenters) uses legacy `.get("indexing",{}).get(...)` dict chains. The
`extra='allow'` silently keeps typo'd config keys.
29     - **Optional module globals** ? `db_instance` and `global_config` are `Optional[...] =
None`, set only in `main()`. Importing `app` without `main()` ? NPE on every endpoint. The mypy
quarantine on `backend.main` and `backend.api.job_worker` exists because of this.
30     - **No-op validator** ? `segmenter_must_be_registered` field validator is a pass-
through; a typo'd segmenter name in `config.json` passes validation and fails at runtime.
31
32     The goal is to fix these systematically, with **zero behavior change** in Phase 1, and a
**strict no-regression gate** (all tests pass, coverage never decreases).
33
34     ---
35
36     ## 2. Decisions locked
37
38     | # | Decision | Source / date | Implication |
39     |---|-----|-----|-----|
40     | **D1** | **Phase 1 = pure refactorings only.** No logic or behavior change. All tests
must pass; coverage must not decrease. | Owner, 2026-06-24 | B2 (Success/Failure contract for
Gemini) is deferred to Phase 2 since it changes return types callers depend on. D3 (typed-config
```

migration), D2 (app-factory), and B5 (segmenter validator) are pure-refactoring-possible and are the first targets. |

41 | ****D2**** | ****First targets: D3 ? B5 ? D2**** in that order (increasing risk). D3 (typed-config migration) is the safest ? fix the validator callers one module at a time. B5 (segmenter validator) is a one-liner. D2 (app-factory) is the largest but still a refactoring. B2 (Success/Failure in Gemini) is the first Phase-2 item. | Owner, 2026-06-24 | Each lands via a separate, small PR branched from `master`; no PR stacking. |

42 | ****D3**** | ****Docs-only PR first.**** This document lands as a single PR with no code changes. The subsequent code PRs reference this doc. | Owner, 2026-06-24 | The doc is approval for the approach; individual code PRs are approval for the implementation. |

43 | ****D4**** | ****Coverage ratchet:**** the CI `coverage` lane must not decrease for any PR in this project. | Convention from `CODE\_AUDIT\_AND\_TEST\_HARDENING.md` | Every code PR must include updated/additive tests. Pure refactorings that remove dead code may increase coverage; a drop is a hard fail. |

44 | ****D5**** | ****Follow the CODE_MAP placement rules.**** No new top-level files without a placement-rule match. Tools go in `tools/`, standalone run-drivers in `scripts/`, eval library code in `backend/eval/`. | `docs/CODE\_MAP.md` §0 | Keeps the repo navigable as it evolves. |

45 | ****D6**** | ****mypy quarantine ratchet:**** when touching a quarantined module, try removing its `ignore\_errors = True` entry from `mypy.ini`. | `mypy.ini` | D2 (app-factory) is the biggest opportunity ? if `backend.main` and `backend.api.job\_worker` can leave quarantine, that's a major milestone. |

46 | ****D7**** | ****Absorb relevant BACKLOG items.**** Deferred hardening items from `docs/BACKLOG.md` that overlap with this audit are absorbed into this project rather than left in BACKLOG. Items that are feature-gaps (not code-cleanup) stay in BACKLOG. | PR #352 review, 2026-06-24 | D2 and O1 were already independently found; D13 and D14 are new absorptions. The train/eval splits guardrail stays in BACKLOG (it needs a collector-side manifest change). |

47

48 ---

49

50 ## 3. Findings register (full audit)

51

52 ### 3a. Bugs & Defects

53

54 | ID | Severity | File(s) | Issue | Phase |

55 |----|-----|-----|-----|-----|

56 | ****B1**** | ****HIGH**** | `backend/pipeline/segmenters/motion.py` | ****Fabricated data.****

Random values for server/receiver/shot_count/events/intensity/avg_speed_kmh persisted as real data. | 2 |

57 | ****B2**** | ****HIGH**** | `backend/providers/gemini.py` | ****Silent zero-segment returns mask API failures.**** `analyze\_video()` returns `([], metrics)` on ANY failure, indistinguishable from "no rallies." | 2 |

58 | ****B3**** | MEDIUM | `backend/pipeline/validators/pts\_continuity.py` | Hardcoded 10.0s gap threshold (comment says 8.0s), not config-driven. | 3 |

59 | ****B4**** | MEDIUM | `backend/pipeline/stitcher/compiler.py::parse\_time` | Unparseable timestamps silently default to 0.0 ? could stitch from time zero. | 3 |

60 | ****B5**** | MEDIUM | `backend/config/models.py::IndexingConfig` |
`segmenter\_must\_be\_registered` validator is a ****no-op pass-through****. Typo'd segmenter passes validation. | ****1**** |

61 | ****B6**** | LOW | `backend/pipeline/validators/resolution\_fps.py` | Hard-fails if
`format\_check` didn't run first ? ordering is an import-time side-effect. | 3 |

62 | ****B7**** | LOW | `backend/eval/calibrate\_wasb.py::load\_golden\_gts` | Silently skips bad rows (opposite of `annotations.load\_annotations` which RAISES). | 3 |

63 | ****B8**** | LOW | `backend/eval/regression.py` | No baseline ? `regressed=False` (silent pass). Library function diverges from `run\_regression.py` exit 3. | 3 |

64

65 ### 3b. Technical Debt / Design Smells

66

67 | ID | Severity | Area | Issue | Phase |

68 |----|-----|-----|-----|-----|

69 | ****D1**** | HIGH | `backend/main.py` | ****Massive god-module**** (~1450 lines). CLI, API, worker orchestration, upload, debug-clip, trim, reingest all in one file. | 2 |

70 | ****D2**** | HIGH | `backend/main.py` + `backend/api/job\_worker.py` | ****Optional module globals**** (`db\_instance`, `global\_config`). NPE-prone; mypy-quarantined. | ****1**** |

71 | ****D3**** | HIGH | `backend/config/models.py` | ****Dual config access.**** Typed Pydantic model + legacy `.get()` dict shim. `extra='allow'` silently keeps typo'd keys. | ****1**** |

```

72      | **D4** | MEDIUM | `backend/config/loader.py` | Shallow top-level merge: a
`config.json` section REPLACES the defaults dict for that section ? new fields missed on
upgrade. | 2 |
73      | **D5** | MEDIUM | `backend/main.py` | `print()` used in production backend code (~20
calls in `--debug-clip`). Bypasses logging. | 3 |
74      | **D6** | MEDIUM | `backend/eval/` | Duplicate "annotations" modules:
`backend.eval.annotations` vs `backend.annotations`. Different schemas, different error
handling. | 3 |
75      | **D7** | MEDIUM | Multiple | Hardcoded tuning constants duplicated by copy (BASELINE,
TUNED, padding). | 3 |
76      | **D8** | MEDIUM | `backend/pipeline/segmenters/base.py` | Registry instantiates
`cls(None, {})` to read `.name` ? `__init__` must not touch `self.db`/`self.config`. Implicit
contract. | 3 |
77      | **D9** | LOW | `backend/pipeline/validators/__init__.py` | Import mutates global
registry as side-effect. Removing import = de-registering validator. | 3 |
78      | **D10** | LOW | `backend/reporting/spec.yaml` + `harvest.py` | Tight coupling via
magic `when.path` strings. Rename a detail ? rule silently stops firing. | 3 |
79      | **D11** | LOW | `backend/pipeline/stitcher/compiler.py` | Two-stage codec contract is
implicit. Change a preset ? concat breaks. | 3 |
80      | **D12** | LOW | `tools/generate_highlights.py` | Uses `sys.path.insert(0, ...)` hack
instead of proper package install. | 3 |
81      | **D13** | MEDIUM | `requirements.txt`, `pyproject.toml` | **Absorbed from BACKLOG.**
`supervision` ByteTrack is deprecated as of 0.28.0 and **removed in 0.30.0**. The `<0.30.0` pin
in `requirements.txt` and `pyproject.toml` is a stopgap ? migration to the standalone `trackers`
package is needed before the pin can be lifted. | 2 |
82      | **D14** | LOW | `tests/` directory | **Absorbed from BACKLOG.** 90+ test files in a
flat `tests/` directory. Owner wants reorganization by subsystem (e.g. `tests/eval/`,
`tests/audio/`) to reduce fragmentation. The exact grouping metric is the owner's design choice.
| 3 |
83
84      ### 3c. Operational Risk
85
86      | ID | Severity | Area | Issue | Phase |
87      |----|-----|-----|-----|-----|
88      | **O1** | HIGH | `backend/providers/gemini.py` | **Orphaned remote files on process
death.** No GC for uploaded videos on Gemini File API. | 2 |
89      | **O2** | MEDIUM | `backend/main.py` | Single `ThreadPoolExecutor(max_workers=1)` ? one
stuck job blocks all. | 2 |
90      | **O3** | MEDIUM | `backend/infrastructure/database.py` | SQLite WAL + busy_timeout but
concurrent writers (CLI + server) risk `SQLITE_BUSY`. | 3 |
91      | **O4** | LOW | `backend/pipeline/detectors/tracknet_runner.py` | `weights_path=''` ?
silent zero results. No startup weights-exist validation. | 3 |
92      | **O5** | LOW | `backend/utils/single_instance.py` | Windows byte-range locks may not
work on network drives (Google Drive, SMB). | 3 |
93
94      ### 3d. Quick Wins (Low Risk, High Value)
95
96      | ID | Area | Issue | Phase |
97      |----|-----|-----|-----|
98      | **Q1** | `backend/config/models.py` | Fix the no-op `segmenter_must_be_registered`
validator (same as B5). | **1** |
99      | **Q2** | `backend/pipeline/segmenters/motion.py` | Fix return type annotation:
`List[Dict]` ? `Result[List[Dict[str, Any]]]`. | 1 |
100     | **Q3** | `run_eval.py`, `run_regression.py`, `scripts/` | Deduplicate
`_default_db_path()` helper into a single shared utility. | 1 |
101     | **Q4** | `backend/config/loader.py` | Cache builtin defaults dict at module level
(avoid double-instantiation of `AppConfig`). | 1 |
102     | **Q5** | `requirements.txt` vs `pyproject.toml` | Add comment explaining why
`torch`/`torchvision` are in requirements.txt but not `pyproject.toml`. | 1 |
103     | **Q6** | `ruff.toml` | Enable `"I" (isort) rule ? auto-fixable, consistent import
ordering. | 1 |
104     | **Q7** | `mypy.ini` | Type-clean `motion.py` (small, legacy) as a ratchet exercise. |
2 |
105     | **Q8** | `backend/main.py` | Add comment explaining lazy `import cv2` in `--debug-
clip` (cv2 not in LIGHT tier). | 1 |

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106
107     ### 3e. Absorbed from BACKLOG
108
109     These findings were already recorded in `docs/BACKLOG.md` as deferred items from the
2026-06-10 hardening sweep. They are absorbed into this project for execution:
110
111     | My ID | BACKLOG item | BACKLOG status | Absorbed as |
112     |-----|-----|-----|-----|
113     | **D2** | "App-factory / lifespan + lazy torch-cv2 imports" | ? ? "Belongs with the
async-job slice (#16)" ? #16 is now done; unblocked | Phase-1 target (P5) |
114     | **D1** | "WSL temp/frame/cache + Gemini remote-file GC" | ? ? "staged videos, PNG
frame dumps, `_wasbcache`, and orphaned Gemini uploads are never cleaned" | Phase-2 target. The
WSL half is partially handled by `keep_frames=false`; the Gemini half is new. |
115     | **D13** | "`supervision` ByteTrack deprecation" | ? ? pinned `<0.30.0` as stopgap;
migration to `trackers` package needed | New finding in this doc (see §3b) |
116     | **D14** | "`tests/` reorganization to reduce fragmentation" | ? ? owner's design
choice on grouping metric needed first | New finding in this doc (see §3b) |
117
118     **Not absorbed** (kept in BACKLOG as feature-gap, not code-cleanup):
119     - ? **Train/eval separation guardrails** ? `splits.py` exists but needs collector-side
manifest change; this is a cross-repo feature, not a code-cleanup item.
120
121     ---
122
123     ## 4. Design / architecture ? Phase 1 approach
124
125     ### 4a. D3 ? Typed-config migration (first target)
126
127     **What:** Convert all call sites that use `config.get("indexing",{}).get(...)` to typed
`AppConfig` attribute access.
128
129     **Reuse map:**
130     - `backend/config/models.py` ? `AppConfig`, `IndexingConfig`, `ValidationConfig` (single
source of truth for defaults + types)
131     - `backend/config/__init__.py` ? re-exports `load_config`, `AppConfig`
132     - `backend/pipeline/validators/*.py` ? 6 validators that consume `config: Dict[str,
Any]`
133     - `backend/pipeline/segmenters/motion.py` ? uses `self.config.get("indexing",{})`
134     - `backend/reporting/__init__.py` ? already accepts both typed and dict; normalizes via
`model_dump()`
135
136     **Plan:** Convert one module at a time, bottom-up. Validators first (they receive config
as a dict, change the contract to `AppConfig`), then segmenters, then remove the `.get()` compat
shim and flip to `extra='forbid'`.
137
138     **Test strategy:** Every module has existing tests. The config models have
`test_config.py` that pins defaults. No new test behavior needed ? existing tests should pass
byte-identically.
139
140     ### 4b. B5 ? Segmenter validator fix (second target)
141
142     **What:** Make `segmenter_must_be_registered` actually validate against
`segmenter_registry`.
143
144     **File:** `backend/config/models.py`, field validator on
`IndexingConfig.default_segmenter`.
145
146     **Risk:** Near-zero. A typo'd segmenter in `config.json` that currently "works" (passes
validation, fails at runtime) would now fail at startup. This is the DESIRED behavior ? catching
config errors early.
147
148     ### 4c. D2 ? App-factory refactor (third target, largest)
149
150     **What:** Replace `db_instance: Optional[Database] = None` + `global_config:
Optional[AppConfig] = None` module globals with a `create_app(config) -> FastAPI` factory that

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injects state via `app.state`.
151
152     **Reuse map:**
153     - `backend/main.py` ? FastAPI app definition, all endpoints
154     - `backend/api/job_worker.py` ? resolves `db_instance`/`global_config` through `import
backend.main as _main`
155     - `tests/test_api_endpoints.py` ? already uses `TestClient`; would switch to
`create_app(test_config)`
156     - `tests/test_async_jobs.py` ? already monkeypatches `backend.main`
157
158     **Plan:** This is the highest-risk Phase-1 item but still a pure refactoring. The key
insight: Starlette's `app.state` is the standard way to attach shared state; FastAPI's
dependency injection can access it. Every endpoint that accesses `db_instance`/`global_config`
would use `request.app.state.db` / `request.app.state.config` instead. The worker would receive
these explicitly rather than resolving through the module object.
159
160     **Test strategy:** All existing tests must pass byte-identically. The `TestClient` usage
changes from `from backend.main import app` to `from backend.main import create_app; app =
create_app(test_config)`.
161
162     ---
163
164     ## 5. Honest limits ? what Phase 1 does NOT do
165
166     - **Does NOT change any behavior.** The Gemini provider still returns `[]` on failure;
the PTS validator still has a hardcoded 10.0s gap.
167     - **Does NOT fix B2 (Success/Failure in Gemini).** That changes the return contract
callers depend on ? deferred to Phase 2.
168     - **Does NOT split `main.py` (D1).** The god-module split is deferred ? D2 (app-factory)
is a necessary precursor.
169     - **Does NOT address any operational risks (O1?O5).** Those need behavioral changes.
170     - **A green suite after Phase 1 does NOT mean the code is "clean."** It means the
structural debt targeted by D2/D3/B5 is resolved. The remaining 28 items still need attention.
171
172     ---
173
174     ## 6. Deliverables & progress tracker ? **source of truth**
175
176     Legend: ? Todo . ? In progress . ? Done . ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking.
177
178     | ID | Deliverable | Depends on | Gated? | Status | PR |
179     |----|-----|-----|-----|-----|-----|
180     | **P0** | **Docs-only PR ? this project doc** + index in `docs/README.md` | ? | Owner
approval | ? | [#352](https://github.com/Khelsutra/badminton-highlight-indexer/pull/352) |
181     | **P1** | **WASB cache test fix (Windows)** ? 2 pre-existing failures | ? | No | ? |
[#353](https://github.com/Khelsutra/badminton-highlight-indexer/pull/353) |
182     | **P4** | **B5: Fix `segmenter_must_be_registered` validator** | P0 | No | ? |
[#354](https://github.com/Khelsutra/badminton-highlight-indexer/pull/354) |
183     | **P1** | **D3-a: Convert validators to typed `AppConfig`** (6 files) | P0 | No | ? |
[#355](https://github.com/Khelsutra/badminton-highlight-indexer/pull/355) |
184     | **P6a** | **Q3: Deduplicate `_default_db_path`** | P0 | No | ? |
[#356](https://github.com/Khelsutra/badminton-highlight-indexer/pull/356) |
185     | **P6b** | **Q4: Cache builtin defaults** | P0 | No | ? |
[#358](https://github.com/Khelsutra/badminton-highlight-indexer/pull/358) |
186     | **P6c** | **Q5: requirements.txt torch comment + Q6: isort (I) + Q8: cv2 comment** |
P0 | No | ? | [#359](https://github.com/Khelsutra/badminton-highlight-indexer/pull/359) |
[#360](https://github.com/Khelsutra/badminton-highlight-indexer/pull/360) |
187     | **T1** | **Tier 1: Q2 (motion return type) + D11 (codec docstring)** | P0 | No | ? |
[#362](https://github.com/Khelsutra/badminton-highlight-indexer/pull/362) |
188     | **T2** | **Tier 2: B6 (resolution_fps fallback + warning log)** | P0 | No | ? |
[#363](https://github.com/Khelsutra/badminton-highlight-indexer/pull/363) |
189     | **T3** | **Tier 3+4: D12 (sys.path hack) + B3 (PTS gap config) + O5 (lock note) + D7
(centralize TUNED)** | P0 | No | ? | [#364](https://github.com/Khelsutra/badminton-highlight-

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indexer/pull/364) |
190 | **P6d** | --D7: Centralize hardcoded tuning constants~~ ? folded into T3 | P0 | No | ?
| [#364](https://github.com/Khelsutra/badminton-highlight-indexer/pull/364) |
191 | **P2** | **D3-b: Convert segmenters to typed `AppConfig`** (15 files: 5 segmenters + 4
detectors + worker + eval + tools) | P1 | No | ? |
[#366](https://github.com/Khelsutra/badminton-highlight-indexer/pull/366) |
192 | **P3** | **D3-c: Remove `.get()` compat shim; flip `extra='forbid'`** (18 files,
+87/-169) | P2 | No | ? | [#367](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/367) |
193 | **P5** | **D2: App-factory refactor** ? replace module-level Optional globals with
`create_app(config, db) ? FastAPI` | P3, P4 | No | ? |
[#370](https://github.com/Khelsutra/badminton-highlight-indexer/pull/370) |
194 | **P7** | **Phase-1 verification:** full suite green, coverage ? 84%, mypy ratchet |
P1?P6 | No | ? | 1463/1463 pass (#371); `backend.api.job_worker` lifted from mypy quarantine |
195 | ? | ? | ? | ? | ? | ? | ? |
196 | **P8** | **B2: Error-flagging in Gemini provider metrics** (Phase 2) | P7 | No | ? |
[#373](https://github.com/Khelsutra/badminton-highlight-indexer/pull/373) |
197 | **P9** | **B1: Remove fabricated data from motion segmenter** ? random
server/receiver/shot_count/speed ? `None`; events dropped | P7 | No | ? |
[#385](https://github.com/Khelsutra/badminton-highlight-indexer/pull/385) |
198 | **P10** | **D1: Split `main.py` god-module** (~1450 lines ? smaller modules) | P7 | No
| ? | ? |
199 | **P11** | **O1: Gemini remote-file GC** ? per-call `_safe_delete` (#373) + startup
orphan sweep scoped to our `UPLOAD_TAG`-tagged files, grace-gated, HTTP-timeout-bounded | P7 |
No | ? | [#380](https://github.com/Khelsutra/badminton-highlight-indexer/pull/380) |
200 | **P12** | **D13: ByteTrack deprecation** ? migrated `sv.ByteTrack` ?
`trackers.ByteTrackTracker` (git dep; PyPI wheel broken) | P7 | No | ? |
[#386](https://github.com/Khelsutra/badminton-highlight-indexer/pull/386) |
201 | **P13** | **D4: Deep config merge** ? section-level replace ? recursive merge | P7 |
No | ? | [#373](https://github.com/Khelsutra/badminton-highlight-indexer/pull/373) |
202 | **P14** | **O2: Single-executor risk** ? documented with mitigations in job_worker.py
| P7 | No | ? | [#374](https://github.com/Khelsutra/badminton-highlight-indexer/pull/374) |
203 | **P15** | **Q7: Type-clean `motion.py`** ? ratchet exercise | P7 | No | ? |
[#373](https://github.com/Khelsutra/badminton-highlight-indexer/pull/373) |
204 | **P17** | **B2 (cont.): finish the `Success`/`Failure` return contract** ? the 4
callers that dropped provider error metrics (`trajectory_hybrid`, `yolo_hybrid`,
`gemini_refine.refine_window`, `gemini_refine.full_video_rallies`) now surface
`metrics["error"]` loudly instead of silently treating a failure as "0 rallies". (Full
`Success`/`Failure` return-type migration of the hybrids is follow-up.) | P8 | No | ? |
[#378](https://github.com/Khelsutra/badminton-highlight-indexer/pull/378) |
205 | **P16** | Phase 3 items (B4/B7/B8/D5/D6/D8-D10/D14/O3/O4) | P15 | No | ? | ? |
206
207 ---
208
209 ## 7. Open questions
210
211 1. **Q: Should D1 (split `main.py`) be Phase 1 or Phase 2?` ? Leaning Phase 2: D2 (app-
factory) is a natural precursor, and D1 is lower risk after the globals are gone. Owner to
confirm.
212 2. --Q: Should Phase 1 include enabling `I` (isort) in ruff.toml?~~ ? ? Shipped in
#360.
213
214 ---
215
216 ## 8. Definition of done
217
218 - [x] P0: This doc approved and merged to `master`
219 - [x] P1?P3: Typed-config migration complete ? no `.get()` dict access remains in
backend code; `extra='forbid'` on `AppConfig`
220 - [x] P4: `segmenter_must_be_registered` actually validates against the registry
221 - [x] P5: `create_app()` injects shared state via `app.state` (handlers read
`request.app.state`); `backend.api.job_worker` removed from mypy quarantine. (The Optional
module globals `db_instance`/`global_config` are **retained as a transitional back-compat shim**
and `backend.main` stays mypy-quarantined ? full removal is follow-up, see D1/P10.)
222 - [x] P6: Quick wins Q1?Q6, Q8 landed

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223 - [x] P7: Full test suite ? 1463/1463 pass (#371); `ruff` + `mypy` clean;
`backend.api.job_worker` lifted from quarantine
224 - [x] All PRs are single-purpose, branched from `master`, with descriptive commit
messages
225 - [ ] `docs/README.md` updated to reflect the project's terminal state
226 - [ ] Final status flipped to `DONE` and this doc moved to
`docs/archives/past_projects/`
227
228 ---
229
230 ## 8.1 ? Next items ? Phase 2 pick-up (prioritized)
231
232 All items below are Claude-doable, small-PR-friendly, and unblocked. Pick up in
order:
233
234 ### Immediate (HIGH impact, low risk)
235
236 | # | ID | What | Why |
237 |---|---|---|---|
238 | ~1~ | ~**B2 / P17**~ | ~**Finish** the `Success`/`Failure` return contract~ ?
?? DONE (this PR). The 3 callers that dropped provider error metrics now surface
`metrics["error"]` loudly. Follow-up remains: migrate the hybrid segmenters' `process_video`
return type from bare `List[Dict]` to the ABC's `Result[...]` (`Success`/`Failure`) ? currently
suppressed with `type: ignore[override]` in `trajectory_hybrid.py` (motion.py is already fully
migrated). |
239 | ~2~ | ~**01 / P11**~ | ~Gemini remote-file GC~ ? ?? DONE (#380). Per-call
`_safe_delete` (#373) + startup orphan sweep scoped to `UPLOAD_TAG`-tagged files, grace-gated,
HTTP-timeout-bounded. |
240
241 ### Near-term (MEDIUM impact)
242
243 | # | ID | What | Why |
244 |---|---|---|---|
245 | ~3~ | ~**B1 / P9**~ | ~Remove fabricated data from `motion.py`~ ? ?? DONE
(#385). Random server/receiver/shot_count/speed ? `None`; events dropped. |
246 | ~4~ | ~**D13 / P12**~ | ~ByteTrack deprecation migration~ ? ?? DONE.
`sv.ByteTrack` ? `trackers.ByteTrackTracker` (git dep; PyPI wheel broken ? filed Roboflow
issue). |
247 | 5 | D1 / P10 | Split `main.py` god-module | ~1450 lines. D2 (app-factory) was the
necessary precursor ? now done. Extract: CLI handlers ? `backend/cli/`, upload routes ? already
in `backend/api/uploads.py`, debug helpers ? `scripts/.` |
248
249 ### Later (deferred)
250
251 | # | ID | What |
252 |---|---|---|
253 | 6+ | P16 | B4, B7, B8, D5, D6, D8?D10, D14, O3, O4 ? Phase 3 items |
254
255 > Removed from this list ? already shipped (see the $6 tracker); they were stale
entries left from before Round 3/4 landed: D4/P13 deep config merge (#373 ?
`loader.py::_deep_merge`), O2/P14 single-executor risk documented (#374 ? `job_worker.py`),
Q7/P15 `motion.py` type-cleaned (#373 ? no longer in the `mypy.ini` quarantine list).
256
257 ---
258
259 ## 9. References
260
261 - Audit source: 33 findings from a deep whole-repo read on 2026-06-24, plus 2
absorbed from `docs/BACKLOG.md` (D13, D14) for a total of 35. Two additional BACKLOG items (D2,
O1) were already independently found and are cross-referenced in $3e.
262 - Prior hardening sweep: `docs/CODE_AUDIT_AND_TEST_HARDENING.md` (completed
2026-06-14) ? the previous audit-driven hardening round. This project is the next wave.
263 - BACKLOG.md `docs/BACKLOG.md` ? deferred items from the 2026-06-10 sweep; $3e maps
the absorbed items.
264 - CODE_MAP.md `docs/CODE_MAP.md` ? placement rules, gotchas, and the cross-cutting

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footguns list.
265 - **mypy.ini** ? quarantine list and the ratchet instruction.
266 - **ruff.toml** ? lint rules; `"I"` (isort) deferred.
267 - **PROJECT_DOC_TEMPLATE.md** `docs/PROJECT_DOC_TEMPLATE.md` ? the template this doc
follows.
268
269 ---
270
271 ### Changelog
272 - `2026-06-24` ? Initial doc created from 33-item deep audit.
273 - `2026-06-24` ? Absorbed 2 findings from `docs/BACKLOG.md` (D13, D14). Cross-referenced
D2 and O1 in §3e.
274 - `2026-06-24` ? Phase 1 begun. P0 (#352) ? #353 ? P4/B5 (#354) ? P1/D3-a (#355) ?
P6a/Q3 (#356). Status ? IN PROGRESS.
275 - `2026-06-24` ? **Round 2.** Quick wins (#358 Q4, #359 Q5, #360 Q6 isort, #361 Q8) ?
Tier 1 (#362 Q2+D11) ? Tier 2 (#363 B6) ? Tier 3+4 (#364 D12+B3+O5). 11 items shipped total.
276 - `2026-06-24` ? **Round 3.** P2/D3-b segmenters (#366) + P3/D3-c extra='forbid' (#367).
18 items shipped.
277 - `2026-06-24` ? **Round 4 (final).** P5/D2 app-factory (#370) + test fixes (#371).
Status ? PHASE 1 COMPLETE. Suite 1463/1463 green. §8.1 added with prioritized Phase 2 pick-up.
278 - `2026-06-25` ? **§8.1 reconciled with the §6 source-of-truth (docs-only).** Removed
three already-shipped items that §8.1 still listed as deferred future work ? D4/P13, O2/P14,
Q7/P15 (all merged in #373/#374) ? and split the unfinished half of B2 out of the done P8 into a
new **P17** (the provider return contract + the `trajectory_hybrid`/`gemini_refine` callers that
still drop `metrics["error"]`). Grounded against `loader.py::_deep_merge`, `mypy.ini`,
`backend/results.py`, and `backend/providers/gemini.py` + its callers. No code change.
279 - `2026-06-25` ? **Accuracy fixes (docs-only).** (1) §8 DoD P5 said the module globals
were "replaced with `app.state`"; corrected ? `create_app()` *added* `app.state` and lifted
`job_worker` from quarantine, but `db_instance`/`global_config` remain as a transitional shim
and `backend.main` is still mypy-quarantined (`backend/main.py:264`, `mypy.ini`). (2) `main.py`
size "~1300" ? "1450" (now 1448 lines) in §3b/§6/§8.1.
280 - `2026-06-25` ? **P17 done (B2 finish).** The 4 callers that dropped provider error
metrics (`trajectory_hybrid.py`, `yolo_hybrid.py`, `gemini_refine.refine_window`,
`gemini_refine.full_video_rallies`) now check `metrics["error"]` and log loudly instead of
silently treating a provider failure as "0 rallies" ? matching the pattern the `gemini`
segmenter already uses. (`yolo_hybrid` was caught during PR #378 review ? it bound `metrics` but
never read it.) Suite 1484/1484 green; ruff + mypy clean. Follow-up: migrate the hybrids'
`process_video` return type from bare `List[Dict]` to the ABC `Result[...]` contract (motion.py
is already migrated; hybrids suppressed with `type: ignore[override]` for now).
281 - `2026-06-25` ? **P9 done (B1 finish).** Removed fabricated `random` data from the
motion segmenter ? server/receiver/shot_count/avg_speed_kmh ? `None`, serve/smash/foul/winner
events dropped entirely (motion was the only `add_event` caller). PR #385 review caught a real
regression: `run_process_and_stitch.py` crashed on `None` >= `min_shot_count` (the `get(?, 0)`
default no longer applied) ? fixed with a `None`-guard matching the API path. Suite 1492/1492
green; ruff + mypy clean.
282 - `2026-06-26` ? **P12 CI fix (#386).** The D13/P12 PR failed the `full-suite` lane with
`ModuleNotFoundError: No module named 'trackers'`. Two root causes: (1) the two new tests hard-
imported the standalone `trackers` package, but it's a git-only dep (PyPI wheel is metadata-only
? roboflow/trackers#474) deliberately omitted from `pyproject.toml`, and the lane installs via
`pip install -e .[dev]`; (2) the git ref `@v2.5.0` was wrong ? roboflow/trackers uses bare
version tags (`2.5.0`), so pip failed with `pathspec 'v2.5.0' did not match`. Fixes: added a
`pip install trackers @ git+?@2.5.0` step to both `ci.yml` and `nightly.yml` full-suite lanes
(same out-of-band pattern as torch); fixed the tag in `requirements.txt`; and guarded the
package-importing test with `pytest.importorskip("trackers")` (matching the repo's `obsreport`
soft-dep convention). All 10 CI checks green.
283

```

6. user (2026-07-02T02:47:39.862Z)

```

1 # Backlog ? deferred items to revisit
2
3 Single tracker for work we consciously deferred, so nothing gets lost between sessions.
Each item records **what**, **why it's deferred**, and **the trigger to revisit**. Every entry
traces to a real decision/PR/doc (no speculative work).
4

```

5 > Maintained as a version-controlled doc for now. Say the word and I'll mirror any/all of these to ****GitHub issues**** instead (or in addition).

6 > Status legend: ? blocks something · ? do when triggered · ? nice-to-have/cleanup.

7

8 _Last updated: 2026-06-23._

9

10 ---

11

12 ## Batch / declarative orchestration

13 - ? ****One-shot "serving startup-script" for the GPU VM**** ? `create\_gpu\_vm.sh` already injects a metadata `startup-script` (`deploy/gcp/ops-agent-startup.sh`), but it only installs the ****Ops Agent****; standing up the live engine on the VM (`kheksutra/deploy/GPU\_VM\_SERVING.md`) is still manual. Propose a `deploy/gcp/serving-startup.sh` fed via `--metadata-from-file=startup-script=?` (or pasted into the GCP console) that, on boot: installs ffmpeg/cv2 libs ? pulls the engine source (from GCS) + builds the venv ? ***(optional)*** the WASB native env (`gpu\_setup.sh` + weights) ? writes the serving `config.json`/`engine.env` ? installs `cloudflared` with the tunnel token ? starts `kheksutra-engine`. So a GPU serving box comes up ****hands-off**** (reuses the `deploy/gcp/nightly\_regression/run\_on\_vm.sh` bootstrap pattern). ****Why deferred ? the design hinge is SECRETS:**** the Gemini key / CF-Access AUD / tunnel token must ****not**** sit in a plaintext startup-script (instance metadata is broadly readable by anyone with `instances.get`); they should come from ****Secret Manager**** (VM SA ? `secretAccessor`) or be injected post-boot. Needs that small secrets-from-metadata design + which config is baked vs fetched ? not just a script (owner asked 2026-06-23). ****Revisit:**** when the GPU-VM serving path is used often enough to automate.

14 - ? ****One YAML manifest drives the whole bucket?reels run**** ? today, converting `gs://` videos into highlight reels is a sequence of API calls (see CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md). Propose a declarative ****batch manifest**** (`pipeline.yaml`): ****inputs**** (bucket prefix/glob or explicit list), ****compute**** target (local/cloud), ****detection**** (segmenter, skip_ai_handoff), per-video ****selection**** rules (all / top-N longest / min duration / a query string), and ****output**** (bucket prefix or local dir + watermark locale) ? driven by one `indexer --batch pipeline.yaml` runner that loops process?query?select?compile?deliver, ****idempotent/resumable**** (skips already-done `video\_id`s), reusing the existing API + the compute-picker + the compute-path provenance. ****Why deferred:**** needs a small design ? the manifest schema, the selection grammar, idempotency/resume, and partial-failure handling ? i.e. a tracked project doc per `PROJECT\_DOC\_TEMPLATE.md`, not a quick add (owner agreed 2026-06-23: ship the runbook doc now, backlog the YAML driver). ****Revisit:**** when hands-off batch conversion is wanted ? start with a design doc.

15

16 ## Owned-model / multisport (source: `OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md`, `NEXT\_STEPS.md`)

17 - ? ****Rename the repo off "badminton"***** ? owner-emphatic "ASAP" action; the project is multisport. Touches remote name + README/CLAUDE.md framing + any hardcoded URLs. ****Revisit:**** owner action, before broad external sharing.

18 - ? ****Quantify cost-to-Gen-2 (the largest open risk)**** ? benchmark real VRAM/token/\$ for the VLM SFT+GRPO on a rented A100/L40S (ms-swift multimodal-GRPO examples use 6?8 GPUs) ****before**** standing up any owned serving stack. ****Revisit:**** before Phase-2 commitment.

19 - ? ****Define the ship-gate F1@tIoU target**** ? "very high P/R" must be operationalized (floor = beat heuristic LOVO 0.480; learned is now 0.626 post metric-fix). ****Revisit:**** at M0.4 harness.

20

21 ## Engineering hardening deferred from the 2026-06-10 audit sweep

22 - ? ****App-factory / lifespan + lazy torch-cv2 imports**** ? `uvicorn backend.main:app` leaves globals None, and importing the API pulls the GPU stack. Belongs with the async-job slice (#16). ****Revisit:**** when the async slice starts.

23 - ? ****WSL temp/frame/cache + Gemini remote-file GC**** ? staged videos, PNG frame dumps, `\_wasbcache`, and orphaned Gemini uploads are never cleaned. Needs a retention policy. ****Revisit:**** before any volume/batch processing.

24 - ? ****shlex-quote WSL config values**** ? ****DONE 2026-06-11 (#97):**** `detectors/base.py::wsl\_tilde\_quote` quotes everything after a leading `~/`, so quoting and tilde expansion coexist; applied to every `~`-capable arg in both runners; `tests/test\_wsl\_tilde\_quoting.py` pins the command shapes (no quoted tilde can ever reach `\_wsl\_bash`). Found live: the naive quoting had silently broken wasb_hybrid staging for the default `~/clips`. Residual for tenancy-era: quote the remaining trusted-config interpolations (repo_dir/conda_env) if config ever becomes per-tenant.

```

25
26 ---
27
28 ## Monetization / licensing
29 *(source: `archives/MONETIZATION_AUDIT.md` §7, ADR-005)*
30
31 - ? Ultralytics Enterprise pricing ? get a quote only if the
torchvision/ByteTrack detector (PR #8) proves insufficient on real footage. Currently not
needed (swap done). Quote-only pricing. Revisit: if detector quality forces reconsidering
YOLOv8.
32 - ? Codec patent thresholds (AVC/HEVC/AAC) ? confirm exact Via-LA (AVC) / Access
Advance (HEVC) royalty rates & free thresholds with legal counsel before relying on the
"small SaaS is under threshold" reading. Mitigation already chosen: emit royalty-free AV1/Opus
output. Revisit: before commercial launch.
33 - ? Per-checkpoint weight/dataset licenses ? if we borrow pretrained weights (YOLOX,
ActionFormer, WASB, TrackNetV3), the code license ? the distributed weights/dataset terms.
Either train our own or confirm each checkpoint. Revisit: when adopting any borrowed
weights.
34 - ? Real GPU $/video & latency ? benchmark the three LLM-step options (offline
ActionFormer vs self-hosted Qwen2.5-VL-7B vs Gemini paid-tier) on the target VM before pricing
the product. Revisit: when choosing the production LLM step.
35
36 ## Detector swap follow-ups
37 *(source: PR #8, ADR-005)*
38
39 - ? supervision ByteTrack deprecation ? sv.ByteTrack is deprecated as of
supervision 0.28.0 and removed in 0.30.0; requirements.txt is pinned <0.30.0 as a
stopgap. Migrate to ByteTrackTracker from the separate trackers package before lifting the
cap. Revisit: before bumping supervision past 0.30.
40 - ? Real-video validation of the swap ? the torchvision + ByteTrack Stage-1 is unit-
tested with mocks but not yet validated on actual footage for detection/tracking quality.
(ADR-004 separately found player pixel-motion is a weak rally proxy on broadcast video, so
expect tuning.) Revisit: when a real test clip + the pipeline are run together.
41 - ? yolo_hybrid naming ? the registry key, yolo_* config keys, and yolo-
hybrid-* DB detector tags keep the "yolo" name for back-compat despite no YOLO remaining.
Optional rename. Revisit: if/when a clean break is worth the migration.
42
43 ## Golden-set subproject
44 *(source: `docs/GOLDEN_SET_IMPLEMENTATION.md`, `docs/GOLDEN_SET_VENDORING.md`)*
45
46 - ? Annotation guideline ? DONE (2026-06-02): authored as the Roora brief +
illustrated guide in the collector repo docs/annotation/ ? rally CSV schema; ignore dead time
/ include every rally (incl. faults & lets) / shot = one contact; ending_reason
(forced/unforced-error taxonomy, shared across net-separated sports); singles & doubles for
badminton & pickleball.
47 - ? Internal gold key ? hand-annotate 5?10 licensed clips to near-perfect as the
vendor answer key. Unblocked. Revisit: anytime.
48 - ? GS-3 ? HumanVendorProvider ? concrete adapter for the chosen annotation vendor
(submit/poll/fetch + DPA/secure-link controls). Vendor selected for the initial pilot: Roora
(Bengaluru; iMerit / Taskmonk / TELUS remain the fallback shortlist). Revisit: once the
Roora pilot confirms quality, then build the adapter.
49 - ? GS-4 oracle model ? AutomatedProvider interface is stubbed (this work) with a
deterministic fake. Wire a concrete fast-tier auto-labeler later. Oracle rule: it MUST be
a different model lineage than production, or the regression test is blind to shared failures.
Revisit: when a production LLM step is chosen (pick the oracle's lineage to differ).
50 - ? GS-5 ? regression triggers ? wire regress() into CI (pre-merge gate), on-
demand CLI (the "surprise" case), and a scheduled run; include a deliberate re-baseline path.
Revisit: after GS-4.
51 - ? Vendor pilot (Roora) ? run the paid pilot with Roora as a single-vendor
pilot first (not a multi-vendor bake-off): assess quality-per-rupee against the $4 bar;
escalate to the iMerit/Taskmonk fallback only if it misses. Revisit: the pilot doubles as
calibration once the gold key exists.
52 - ? Phase-1 video collection ? gather the round-1 golden corpus per
docs/GOLDEN_SET_PHASE1_VIDEOS.md: static self-recorded footage (mirror deployment, not
broadcast), 3 sports incl. singles & doubles, across the diversity matrix; ~12 clips/sport with

```

```

**? held out by match**. **Unblocked.**
53     - ? **Train/eval separation guardrails** (was `DEFERRED_HARDENING_PLAN` #13; design
preserved here on archival 2026-06-14) ? train & eval videos live in **separate catalogs**
today, kept apart by convention only; nothing *enforces* that a video used to train the learned
segmenter (`backend/eval/training.py::build_training_set` ? `distill_local`) is absent from the
eval/regression set, so leakage would silently inflate the headline metrics. **Concrete design**
(from the archived DEFERRED plan §2): split unit = **match, never clip** (hold ~? of each sport
as eval, per `GOLDEN_SET_PHASE1_VIDEOS.md`); add a `split: "train"|"eval"` field to the
collector's `curate export` manifest (cross-repo contract, ADR-002); **[DONE ? pure-logic core,
PR #171]** `backend/eval/splits.py` now exists with `load_split(manifest) -> (train_ids,
eval_ids)` keyed by **match id** (not file MD5 ? re-encodes share a match but differ in
`compute_video_id`) + `assert_disjoint(train_ids, eval_ids)` that raises on overlap, with
`tests/test_splits.py` (overlapping ids ? `assert_disjoint` raises). **STILL DEFERRED:** wire
the guard into the training entrypoint (`distill_local`/`training`) and into `run_regression.py`
(baseline computed strictly on the eval split); add the collector's `curate export` manifest
`split` field (above); the integration test (a deliberately-leaked manifest fails the training
entrypoint). Full rationale/risks: [archives/DEFERRED_HARDENING_PLAN-
COMPLETED-2026-06-14.md](archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md) §2.
**Revisit:** dedicated design session with the user (needs a collector-side manifest change).
54
55     ## Golden regression fixtures (video-free, non-attributable regression suite)
56     *(source: `docs/GOLDEN_REGRESSION_FIXTURES.md` §7 tracker / §9 definition-of-done)*
57
58     Core suite **shipped** (M1·M2·M3·MX·P1·P2·P3·DOC ? ? synthetic + 6 de-attributed real
`fx_r01`?`fx_r06` fixtures replay cross-OS, leak-guarded, CONTROL-only). The two items below are
the **gating remainder** of the doc's §9 definition-of-done (it flips to `DONE`/archive only
when **P0 + M1?M4 + P1 + DOC** are ?); 01/02 are explicitly optional. Tracked here so the track
isn't silently treated as "done" while DoD items remain.
59
60     - ? **M4 ? quality-floor gate + first `regression_baseline.json`** *(recommended-next;
unblocked).* Assert per-fixture **and** mean F1 ? a committed floor
(`tests/test_golden_quality_floor.py` + `eval_baselines/regression_baseline.json`). *First
move:* seed the per-slug floor from the F1 already frozen in each `fx_*/expected.json` `score`
(real fixtures sit at **0.0?0.59**, so floor = `current ? tolerance`, not a global bar).
**Gotchas:** record a **windowing fingerprint** in the baseline (else a SERVED retune silently
passes); keep it **CONTROL-only** (learned/numpy-BLAS is non-deterministic across CPUs); add it
to the `golden-crossos` CI matrix. **Revisit:** anytime ? it's the cheapest step to flip the
track to DONE.
61     - ? **P0 remainder ? functional video-name rename** *(deferred per owner safe-scope,
2026-06-16).* Rename the functional names (`eval_baselines/*.json` keys,
`rally_seg_eval.MULTICOURT_CLIPS`, quality-log names) ? opaque ids. *Why it's hard:* the keys
feed `training/gen0/harness.py`'s LOVO floor matched to the **gitignored
`output/human_lovo_manifest.json` + data files** ? rename them together or the floor breaks;
also collides with the hot quality track. *First move:* put the names in a gitignored
`.leakscan_private` and run `python -m tools.leak_scan --names .leakscan_private` to enumerate
every occurrence, then plan the coordinated rename. **Revisit:** before any open-sourcing /
public commit.
62     - ? **01 ? protobuf + deterministic serialization** *(optional).* `rally.fixture.v1`
proto3 + codegen + `protoc`/`git diff --exit-code` guard. **Revisit:** only if cross-OS **byte-
stable** snapshots are wanted (JSON parse-compare suffices today).
63     - ? **02 ? Tier-1 key-gated path** *(optional).* `scripts/fetch_golden.py` (OneDrive
token) and/or SOPS+age, `RALLY_GOLDEN_DIR` wired, Tier-1 CI gated on a secret (never on fork
runners). **Revisit:** only if sharing person/biometric streams (Tier-1) becomes necessary.
64
65     ## Recording best practices
66     *(source: `docs/VIDEO_RECORDING_GUIDELINES.md`)*
67
68     - ? **Validate recording hypotheses** ? empirically test which capture conditions
(resolution, fps, camera angle/height, lighting) actually move precision/recall, then promote
the winners into user-facing "how to record" guidance. Only the static-camera rule is currently
evidence-backed (ADR-004). **Revisit:** once a controlled corpus + the eval loop can measure
per-condition deltas.
69
70     ## Code & test organization
71     *(source: owner review comments on PR #35, 2026-06-07)*

```


72

73 - ? **tests/` reorganization to reduce fragmentation** ? owner wants to isolate test components on a grouping

74 metric (by subsystem/component) so `tests/` is less fragmented, mirroring the new `backend/pipeline/audio/` +

75 `backend/eval/audio/` layout (e.g. audio tests grouped under `tests/audio/`, eval tests under `tests/eval/`).

76 The exact grouping **metric is the owner's design choice** (consolidate small files vs. add per-area dirs ? the

77 "lesser fragmentation" goal slightly tensions with adding dirs), so this is left as a **dedicated refactor PR**,

78 not guessed here. **Revisit:** when the owner specifies the grouping scheme. (The `backend/eval/audio/` half of

79 the request ? comment 1 ? was done in PR #35.)

80

81 **## Commercial Readiness & Low-Regret Moves ? Approved Decisions Checkpoint (2026-06-07)**

82

83 ***(source: Commercial Readiness Review + structure/sequencing discussion with Grok as second-in-command)***

84

85 - ? **Docs-only checkpoint PR** ? This update + the refreshed `docs/archives/past\_projects/low-regret-moves/LOW\_REGRET\_MOVES\_PLAN.md` (since archived) formally record the approved path before any code changes. The initial review docs (PR #37) introduced the findings; this entry locks the execution decisions.

86 - ? **Approved directory structure** ? Adopt a layered approach under `backend/` for professionalism and future growth (no flat dumping of new modules, no explosion of 15+ single-file dirs):

87 - `backend/config/` (package ? start with this)

88 - `backend/api/` (FastAPI app, routes, static UI)

89 - `backend/pipeline/` (validators + segmenters + stitcher + audio + detectors)

90 - `backend/infrastructure/` (database + external adapters)

91 - Keep `backend/eval/` prominent as a first-class concern.

92 - Logical grouping for providers, sports, annotations, utils, and a small `core/` only when needed.

93 - ? **Sequencing (Config first, then atomic refactor)** ? Implement Deliverable 1 (Config) **first**, introducing the `backend/config/` package immediately. Perform the full directory structure refactor as a **separate high-pri atomic PR** afterwards. This minimizes conflicts, keeps PRs focused, and gives Claude a clean window with no concurrent moves.

94 - ? **Deliverable 1 (Config)** ? `backend/config/` package (`models.py` + `loader.py`) instead of a single `config.py`. Typed `AppConfig`, central loader with built-in defaults + file override, `extra="allow"` for migration. Update `setup.py` + `main.py`. Add tests. Land this before the layout move.

95 - ? **Subsequent items** ? See the refreshed `docs/archives/past\_projects/low-regret-moves/LOW\_REGRET\_MOVES\_PLAN.md` (archived) for the full ordered list (Error contracts, logging, packaging, detector abstraction, etc.). The structure refactor is now explicitly tracked as its own high-pri step right after Config.

96

97 **Status legend reminder:** ? = approved / in progress tracking, = do when triggered.

98

99 **## Observability reports ? shipped; intersections with the low-regret items (2026-06-07)**

100 ***(source: per-video observability feature ? indexer PR #41/#42/#43, collector PR #11/#12; shared lib `obsreport`)***

101

102 - ? **Low-regret items 1 (Config pkg, PR #39) + 2 (directory restructure, PR #40) are DONE.** Remaining = items 3-9 in `docs/archives/past\_projects/low-regret-moves/LOW\_REGRET\_MOVES\_PLAN.md` (archived).

103 - ? **Per-video observability reports shipped in BOTH tools** (a **TECHNICAL** report + a **CUSTOMER-friendly** summary), via the **soft-dependency** lib `obsreport` (github.com/avidullu/sports-obsreport, MIT, v0.1.3). Each run writes `

104 - ? **Item 8 (Input-Quality UX) is LARGELY SUBSUMED** by these reports (validation stage + footgun flags + verdict already surface quality; a clean customer summary already exists). **Do NOT build a parallel mechanism** ? instead **EXTEND** the report: add an audio-readiness stage/flag to `backend/reporting/` (harvest) + the obsreport spec. **Revisit:** when item 8 is picked up.

```

105 - ? **Item 7 (unify hybrid segmenters) should land before the report's real AI-provider
timing.** The report records AI timing as `not_captured`; once `wasb_hybrid`/`tracknet_hybrid`
are unified there is ONE place to thread a recorder callback through `provider.analyze_video` ?
the report. **Revisit:** with item 7 (Claude to add the timing hook there).
106 - ? **Item 3 (Error/Result contract) synergizes with the report.** Today the report
INFERS failures (e.g. `silent_provider_noop` = 0 segments + no API key). A structured
Result/error from providers/segmenters would let the report read the real reason as a first-
class flag. **Revisit:** with item 3.
107 - ? **`obsreport` is a SOFT/commented dep** in `requirements.txt` (works locally via
editable install; pipeline no-ops if absent). **Pin the `git+ssh@v0.1.3` tag** in both repos
once SSH/CI access to the private lib repo exists.
108 - ? **`run_all_tests.py`** added at repo root ? run the full suite (clear PASS/FAIL,
pytest exit code) before sending PRs.
109 - ? **Config-migration follow-ups (DONE ? config & identity consistency pass)** ? the
two typed-config bugs from
110 #39/#40 are fixed: (1) **`/api/compile` now honors `--output-dir`** and (2)
**`output_dir` is a real field on
111 `OutputConfig`** (default `"output"`); the CLI overrides it on the typed model in
`main()`, retiring the
112 write-only `_runtime_overrides` hack. Same pass also unified `video_id` MD5 derivation
into
113 `backend/utils/hashing.py::compute_video_id` and migrated all four `run_*.py` scripts
off raw `json.load` onto
114 `backend.config.load_config`. See `CODE_MAP.md` §2.0/§6 (footguns retired).
115

```

7. user (2026-07-02T02:47:40.179Z)

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1 # Next steps (live) ? refreshed 2026-06-21
2
3 > **How to read:** this top section is the **current cross-track state + prioritized
pending list** (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed
**owned-model & rally-quality track** plan (2026-06-13) is **preserved below, still valid** ? it
is now *one of several* active tracks. Live blow-by-blow: memory `current-state.md`. Doc
lifecycle/hygiene: `docs/DOC_STATUS.md`.
4
5 ## ? Where we are (2026-06-21)
6 **DECOUPLED_COMPUTE delivered on GCP + archived** (#270); **compute = GCP only**
(ADR-013, D0 dropped). **`COMPUTE_DECOUPLED_SERVING`** subproject spun up (#272, **ADR-014**) ?
one pure core + local/cloud profiles; **owner M0 design sign-off pending**.. **Per-video
workspace P0** shipped (#264); **doc-status register + archival due-diligence** in place (#273).
The **khelsutra site + per-rally UI** ship in parallel (sibling track). The **rally-quality /
owned-model** track (below) is the core ML work, gated on **growing the golden corpus**.
7
8 > **? LAUNCH DECISION ? NO-GO (owner, 2026-06-30).** The rally-detection product is
**not** cleared for an external/commercial launch; we stay in **development**. This is the
expected default for the dev phase. Two unblockers gate a future GO: **(1)** set + meet the
**ship-gate target (#172)** ? currently unset, needs corpus growth (and the R2 tolF1 ablation,
item 8); **(2)** resolve the **WASB shuttle-detector weight provenance** (commercial-clean
licensing). P2b (#396) was a do-no-harm quality increment, **not** a launch gate ? merging dev
PRs ? launching. Re-decide only when both unblockers clear.
9
10 ## ? Prioritized pending ? all tracks
11 **P0 ? highest leverage / unblocks the most**
12 0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few
recordings lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-
only** (`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra
`web/` SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable
**service** (M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **(ii)**
the **SPA ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ?
poll `/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290,
default-OFF) to gate alpha tenants, **(iv)** a **compute PICKER in the UI** ("your machine vs
our cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full
record: [[ui-driven-cloud-serving-gap]].
13 0b. **? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable

```

batteries-included app ? `docs/PACKAGING\_PLAN.md`.** The LIGHT-tier engine is bundle-ready: `pip install` ? `indexer --app` serves the **bundled SPA** single-origin, **torch-free**, with app-home state / ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b, P2-launcher/P2a/P2d/P2e). **? Next (fresh focused pushes):** ***(1)** P3 **GPU validation + screencast** (L4, ~\$0.50, DELETE-after, \$100 cap ? `[[gcp-vm-always-delete]]`; ? native-WASB 0-rally tuned-config blocker), ***(2)** P2b-install (uv/script + truly-local default config), ***(3)** P2c first-run wizard (cross-repo khelsutra SPA from `/api/capabilities`), then P3-cloud (Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-app]]`.

14 1. ? **`COMPUTE\_DECOUPLED\_SERVING` M0?M9 LARGELY SHIPPED** (2026-06-21, ADR-014 ratified). M1?M4 (#279/#280/#282/#283), **M5 #286, M6 #287, M8 #288, M9-auth #290** all merged (default-OFF); the **real M7 L4 run** validated the worker (22 rallies, VM deleted, \$0). All \$8 gating Qs locked (D7?D17). **Owner residual** (not blocking Claude's default-OFF code): counsel ToS/DPA (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF team-domain/AUD + ingress-lock; real cloud verification authorized within a **\$100/?10k cap** (D17).

15 2. **[owner, ongoing] Grow the golden corpus** ? multi-court badminton + ?3 matches/sport (TT first). The single highest-leverage move; the quality track *and* full-corpus training feed on it.

16 3. **[owner + Claude + GPU] Full-corpus training (n?5)** ? a real `weights/<ver>/` (the `0.1.0-l4` bundle is n=2 plumbing-only).

17

18 **P1 ? near-term, mostly Claude-doable (default-OFF, green-gated)**

19 4. **[Claude] `COMPUTE\_DECOUPLED\_SERVING` batch ? REMAINING: M11 + M12** (M5/M6/M8/M9-auth ? merged this session). **M11** = `backend/flywheel.py` capture?store?shadow-eval?goldenize plumbing reusing `workspace.py::for\_video` + `backend/eval/experiment.compare` + `backend/eval/promotion.py::promote\_if\_served\_safe`, behind `flywheel.\*`=OFF + a faked `consent\_attested` (? privacy-sensitive ? give it fresh focus). **M12** = `backend/storage/gdrive\_api.py` `GDriveApiStorage(StorageBackend)` + a `gdrive-api://` scheme (regex already allows the hyphen; add to `\_KNOWN` + `\_backend\_class` + the StorageConfig Literal) against an INJECTED fake Drive (mirror `gcs.py` DI); keep the mount-only `gdrive://` untouched. ? **M9-consent cols = a new v7 migration** (M8 took v6). M9-consent ToS/DPA text = owner/counsel.

20 5. **[Claude] Per-video workspace P1?P6** (P1 v6 DB migration ? P2 worker ? P3 per-video routing of `detector/+`tmp/` \*[the `output/chunks_` de-hardcode base = ? done via #282]* ? P4 ? P5 Launch-Report flip ? P6 GPU parity). ? P1's v6 migration coordinates with the M8 v6 cost-ledger migration ? author the version bump once.

21 6. ? **Ops-Agent codified (#291)** ? `deploy/gcp/create\_gpu\_vm.sh` (the one-command VM helper) always bakes the Ops-Agent startup-script; provision.sh grants the APIs+SA roles. PROVEN active on a fresh VM 2026-06-21. ([[gcp-ops-agent-enable]])

22 7. **[owner] Set the ship-gate target (#172)** ? needs corpus growth.

23 8. **[owner-side GPU] R-series: promote-or-reject `R2` toLF1 as the gate** (the ablation) ? needs GPU-extracted golden features.

24

25 **P2 ? later / gated / deferred**

26 9. Rally-quality **P3/P4** (person-fusion LOV0, audio cue) ? needs GPU golden features.

27 9b. ? **DONE (#396):** the R4 density fix (finding B / P2b in `RALLY\_DETECTION\_FIXES\_PLAN.md`) is **promoted to default-ON**. Golden-set A/B at the **SERVED** windowing showed toLF1 0.353?0.364 (?toLF1 +0.010, no regression), so `inactivity\_temporal\_density` default was flipped to True. (Landed flag-gated default-OFF in #394.) Fixed the sparse-tracking quirk where one fast sample after a detection gap held the rally tail open.

28 10. **M0.4 scorer swap** ? TemporalMaxer (per the 2026-06-17 external review).

29 11. **Multisport** (table-tennis first, then pickleball).

30 12. Serving **M8/M9** ? per-video cost ledger, edge auth, DPA/consent (within `COMPUTE\_DECOUPLED\_SERVING`; owner/legal-gated).

31 13. **WASB-C3** weight provenance (gates *sharing* a trained model) ? legal.

32 14. **Rename repo off "badminton"** (owner, BACKLOG).

33 15. Owner housekeeping: retire the shared **DO droplet** + **DO token**; collector **md5** export (collector repo).

34 16. **PMF field validation** (owner-side, business track).

35

36 **Doc-hygiene (tracked in `docs/DOC\_STATUS.md` \$4):** ? #1 per-video reconcile + ? #2 this refresh; **pending:** field/PMF/Roora ? archives + a data-layer `docs/` dir (owner boundary confirm); confirm the COMPLETED top-level stubs; verify `HOSTING\_PLAN`/`UI\_PROPOSAL` currency.

```

37
38 ---
39
40 ## Owned-model & rally-quality track ? detail (2026-06-13 plan; still valid)
41
42 **Updated:** 2026-06-13 (**rally-detection quality track**: fusion + experiment-harness
design docs merged,
43 owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior:
2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
44 ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-
first delivery ? see
45 `docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**;
```

PR-2

```

46 upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload
MD5 byte-identity +
47 compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep
#90?#98 landed** ? ruff/mypy/
48 coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin
collapse into
49 `trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626
re-verified exact).
50 **Authoritative roadmap:**
51 `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-
blow:** memory
52 `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do
next," refreshed each session.
53
54 ## ? Next-session / GPU-box pickup (2026-06-15)
55 **M0 in-container quick wins are COMPLETE:** ship-gate provisional blocker (**#170**),
train/eval split guard (**#171**), the **doc-consistency sweep** (**#174** ? corrected 16
drifted docs vs the code), and the **P3 person-feature fusion litmus machinery is already
present + suite-green** (`backend/eval/fusion_compare.py` + `ablation.py`: person-driven ?F1 /
bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted
data). ? CI is red repo-wide due to a **GitHub Actions minutes/billing outage** (account-level,
not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing
? Actions minutes / spending limit).
56
57 **Immediate next step ? the real P3 LOVO measurement** (this box has a GPU +
`torch+cu124` + `cv2`; videos and trajectory CSVs are present, but the GPU-extracted golden
features + `.rallies.csv` labels are not yet here):
58 1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest
output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-
data\features\2026-06-15-n6" ? produces `*_golden_features.json` for the 6 golden videos. (GPU
+ WSL + videos + trajectory CSVs + `.rallies.csv` labels are owner-side only.)
59 2. Artifacts land in the **shared cloud folder** `C:\Users\avidu\OneDrive\rally-
annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow:
**[`GOLDEN_DATA_SHARING.md`](data_pipeline/GOLDEN_DATA_SHARING.md)**; tracked in **#175**.
60 3. **[CPU dev/agent session]** `python -m backend.eval.fusion_compare --features-dir
<shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir
<shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to
also pass the `skipif` litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`).
61
62 **Also queued (owner-gated / pending input):**
63 - Ship-gate target calibration ? **#172** (owner decision; needs corpus growth).
64 - Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent
will do it).
65 - Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config
bug.
66
67 **Leaving (bulky / M2 ? no start without owner go):** ActionFormer (M0.4), event-index
DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack
deprecation, multisport, repo rename, PMF execution.
68
69 ## Where we are (one paragraph)
70 ADR-008 ratified ("repo split driven by productionization, not isolation"): training

```

lives in-repo (`training/`)

71 behind a CI-enforced import wall; the featurizer is single-sourced
 (`backend/features/rally\_seq.py`, train==serve);

72 the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest
 output/human_lovo_manifest.json --floor

73 eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0
 (LOVO **0.626**, floor passed,

74 sign test 5/6 wins $p=0.109$? honestly not significant, ship=False). The **n=6** owned
 corpus is in the collector

75 (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet
 reclassification). The **Gemini**

76 battery is done (see table) ? the moat is quantified.

77

78 **## The quality table that now drives strategy (IoU 0.5 vs human golden labels)**

79 | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |

80 |---|---|---|

81 | Heuristic (velocity windowing) | 0.657 | 0.157 |

82 | **Gen-0 owned head** (LOVO, n=6) | 0.744 | 0.561 |

83 | Two-stage WASB+Gemini refine | 0.83 | ? |

84 | **Gemini wrapper** (`gemini-flash-latest`) | **0.93** | **0.66** |

85

86 **Reading:** clean footage is commoditized (serve it with Gemini for cents). **Multi-**
 court drops the commodity

87 wrapper to 0.66 ? that is the owned model's ship target (its FPs are background-game
 pickups + dead-time merges,

88 the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-
 growth-sized gap, not an

89 architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG
 tapes (uploads candidate clips

90 only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry
 load-bearing.

91 Baselines tracked in `eval\_baselines/` (heuristic_lovo_n6.json,
 gemini_wrapper_2026-06-10.json).

92

93 **## Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation
 in progress)**

94 **This is the track a desktop/local agent is slated to pick up.** Design docs landed (PR
 #112 + session). All implementation is **owner-gated** (per explicit in the plan); each step =
 ONE PR off `master`.

95

96 **Hardening loop (T1-T5 positive tests + audit + finale) COMPLETE (2026-06-14, PRs
 #156/#157/#160/#161/#163 + #165 + final records).** See archived
 `docs/archives/CODE\_AUDIT\_AND\_TEST\_HARDENING-COMPLETED-2026-06-14.md` (Living Log with every
 pre/post/review) and `docs/archives/HARDENING\_LOOP\_HANDOFF-COMPLETED-2026-06-14.md` (full rules
 + STATE). Top-level now pointer stubs per #165 review. All 5 items + verification + archive
 done; clean slate.

97

98 - **Docs:** `docs/MULTI\_SIGNAL\_FUSION\_PLAN.md` (shuttle + person + audio, one fusion
 layer) .

99 `docs/EXPERIMENT\_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) .

100 `docs/ROORA\_LABELING\_GUIDELINES.md` (v8) . `docs/HOW\_RALLY\_DETECTION\_WORKS.md`
 (2-layer mental model).

101 - **Key finding (no re-investigation needed):** the owner's "held-shuttle counted as a
 rally" bug is **NOT** a missing capability ? `backend/pipeline/detectors/rally\_gate.py` already
 implements the net-crossing **"Gate A"** (`apply\_gate(..., min\_crossings=2)`) that kills
 service-feed/held-shuttle windows, but it is **only** called from eval drivers (and there was no
 `min\_crossings` knob in `IndexingConfig`). **Wiring Gate A into the live segmenter (now via
 fusion_hybrid)** was the single cheapest, highest-confidence quality win. (P2 FusionSegmenter
 specifics: registry `fusion\_hybrid`, **default-OFF**, seeds from substrate, persists
 `net\_crossings` + optional person features, optional served Gate A/B via config knobs;
 adversarially reviewed; suite green. See `tests/test\_served\_gate\_a\_regression.py` for the honest
 finding that on production windowing Gate A is neutral/negative in some cases, hence kept opt-in
 with knob for safety, and the leaderboard revert note.)

102 - **Current status (2026-06-14):** P1 (person cue extraction to reusable
 `PersonDetector`) + P2 (FusionSegmenter + Gate A signal + served handoff gating) + experiment

harness P1 DONE (adversarial review + NO-REGRESSION; suite green). **P3 next (this PR + follow-ups):** learned fusion (person features ? harness `compare` LOVO lift on the 6 golden videos; eager-person-compute optimisation); then P4 audio via hit_detector.

103 - **Pick-up order (each = ONE PR off `master`, owner approval first, flag-gated default-OFF, promote only on measured LOVO):**

104 1. (P2 first-step, largely done via fusion_hybrid) Wire Gate A into production served path + config knob (min_crossings); apply in hybrids at runtime. (Now available opt-in; served neutral on production windowing per litmus ? kept opt-in with knob for safety.)

105 2. P3 first step (this work item): harness `compare` LOVO litmus for person-on variant (players_in_court + two_sided + colocation etc. from `fusion\_features`) on the 6 golden; add the eager compute optimisation note/landing. Pure + harness (testable here).

106 3. P3/P4 continuation: audio cue integration + full learned promotion if lift proven.

107 - **Discipline (from EXPERIMENT_HARNESS.md):** every new cue **flag-gated, default OFF** ; promote to default **only on measured LOVO lift** through `run\_regression.py` (exit 0/1/3); pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.

108 - ? Gate A, person cue logic, harness, and pure fusion paths are testable here; the heavier real-video / golden LOVO / ablation confirmation waits on the GPU-extracted golden features + `rallies.csv` labels, which are not yet present in this checkout.

109 - **External due-diligence review inputs (2026-06-17) ? PICK UP AFTER current ideas are exhausted; full details + citations in [`RALLY\_QUALITY\_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md) §5.6.** An owner-commissioned 31-reference review of the quality report *validated* the methodology and surfaced these **future** levers (research inputs, not active work, not new claims; verify citations before any submission):

110 1. **"First-mile" paper framing (highest-value):** stroke-classifiers (BST/DSTA) + the big datasets (ShuttleSet ~36k strokes, Fine-Badminton ~27k actions) all **assume a pre-trimmed rally clip already exists** ? we solve the ignored prerequisite. Pitch the paper as **"the missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage."**

111 2. **Ground R2 in Action-Spotting / Precise-Event-Spotting** (SoccerNet/TennisSet/T-DEED) ? turns R2 into a defensible publishable contribution.

112 3. **Add domain baselines for any submission:** MonoTrack + BST on ShuttleSet / Fine-Badminton (we have only heuristic + Gemini today).

113 4. **Prefer TemporalMaxer over ActionFormer** for the M0.4 scorer swap (parameter-free max-pool, ~2.8x fewer GMACs, small-N robust) ? see "M0.4 next increment" below.

114 5. **Court polygons ? pose ? the rally-END lever:** masking on-court players can resolve *occluded end-of-rally states* (the #1 remaining gap vs Gemini), not just spectator noise.

115 6. **Name the eval?serve lesson "pipeline distribution skew"** ? a citable paper "lesson."

116

117 **Discipline reminder:** owner approval before starting any owner-gated item; one PR per step; babysit the PR (poll/fix/reply) until merge observed, *then* advance.

118

119 ## Prioritized next steps

120 1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move; every lever below feeds on it.

121 Priority order for new videos: **(a) multi-court badminton** (the target distribution AND where the ship target

122 lives), **(b) ?3 matches per new sport ? table tennis first** (WASB transfers at F1 0.909; pickleball labels are

123 corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c) capture variety per

124 `VIDEO\_RECORDING\_GUIDELINES.md`; **adult sessions only for the training pool** (consent guardrail). Per video:

125 label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ? re-run the Gen-0 harness.

126 The paired sign test needs n: at n=10 with 9 wins, $p > 0.011$? significance is reachable this month.

127 2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480 (necessary), **multi-court

128 reference = wrapper 0.66** (the number that makes the owned model worth serving), clean-footage ceiling = 0.93

129 (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70 on multi-court folds with

130 paired-sign $p < 0.05$ vs heuristic" ? owner to ratify/adjust.

131 3. **M0.4 next increment (\$0/local):** swap the Gen-0 LogReg for a lightweight temporal

```

head (MIT, minutes on the
132     2070S ? NOT the heavyweight Gen-2 stack) inside the existing harness; per-video
threshold calibration (the
133     LOVO-vs-oracle gap is visible per fold in the harness output). **Prefer
TemporalMaxer** (parameter-free
134     max-pool, ~2.8x fewer GMACs, small-N robust) over ActionFormer/ASFormer per the
2026-06-17 external review
135     (`RALLY_QUALITY_RESEARCH.md` $5.6) ? better fit for the cheap/local/fast mandate.
136 4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`);
merged-prototype LOVO across
137     badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-
on-WASB serving stays gated
138     by C3 (provenance multiplies per sport).
139 5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+
the two added questions:
140     "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot
at the warmest 2-3 academies
141     (consent at capture; demo served via the Gemini path or human-polish ? the reel
pattern exists:
142     `output/MahadevpuraSingles_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner
ratified the
143     physical-visit approach; reframed product-feedback-first per owner spec):**
144     `docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md` (the recruiting flyer ? owner fills ?/contact
placeholders and recruits) +
145     `docs/FIELD_VISIT_PLAYBOOK.md` (the hired associate's visit playbook: coach + student
script, demo-after-Q6
146     rule, ground rules, Phase-2 record-and-show-back stretch goal) +
`docs/FIELD_VISIT_ANSWER_SHEET.md` (per-visit
147     capture form incl. demo-reaction + student blocks) + `docs/PMF_FIELD_SIGNALS.md`
(G/A/R anchors + **pre-registered
148     decision rules** ? PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first
visit).
149 6. **Housekeeping:** ~~rotate the Gemini API key~~ **DONE 2026-06-10** (owner-verified
working; confirm old-key
150     deletion in AI Studio); **rename repo off "badminton"** (owner action,
151     in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the
battery completed.
152
153 ## Pointers / artifacts
154 - Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` . Topology: ADR-008
(`docs/archives/decisions/DECISIONS.md`) .
155 Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` . Quality history:
156 `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
157 - Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact
bundle; weights gitignored,
158     manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
159 - Corpus: collector DB (system-of-record, 6 matches/262 rallies) . eval manifest
`output/human_lovo_manifest.json`
160     (gitignored) . durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation
Setup\Collect\Trajectories\` .
161     labels: `...\Collect\Golden Labelled\`.
162 - Gemini ops learnings: serial uploads (`indexing.ai_max_workers=1`) + chunk re-encode
163     (`indexing.ai_chunk_scale_width=1280`) + 6-attempt exponential generate retry ? a
cold/prepaid project burst-throttles
164     with a MISLEADING "credits depleted" 429; ramp gradually.
165 - ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next
milestone. Committed next
166     PLATFORM slice = async job model (single-tenant).
167

```

8. user (2026-07-02T02:47:49.691Z)

```

docs/DECISION_SNAPSHOT_REGRESSION.md:16:## 0. TL;DR
docs/DECISION_SNAPSHOT_REGRESSION.md:19:## 1. Problem & goal
docs/DECISION_SNAPSHOT_REGRESSION.md:26:## 2. Decisions *(D1?D11 **LOCKED** 2026-06-25)*

```

docs/DECISION_SNAPSHOT_REGRESSION.md:42:## 4. Design / architecture
docs/DECISION_SNAPSHOT_REGRESSION.md:53:## 6. Honest limits ? what this does NOT do
docs/DECISION_SNAPSHOT_REGRESSION.md:60:## 7. Deliverables & progress tracker ? **source of truth**
docs/DECISION_SNAPSHOT_REGRESSION.md:75:## 8. Open questions ? owner
docs/DECISION_SNAPSHOT_REGRESSION.md:81:## 9. Definition of done
docs/DECISION_SNAPSHOT_REGRESSION.md:89:## 10. References
docs/GOLDEN_REGRESSION_FIXTURES.md:11:## 0. TL;DR
docs/GOLDEN_REGRESSION_FIXTURES.md:26:## 1. Problem & goal
docs/GOLDEN_REGRESSION_FIXTURES.md:38:## 2. Decisions locked
docs/GOLDEN_REGRESSION_FIXTURES.md:52:## 3. Legal foundation (researched ? gates on counsel, §8)
docs/GOLDEN_REGRESSION_FIXTURES.md:76:## 4. Architecture (design)
docs/GOLDEN_REGRESSION_FIXTURES.md:117:## 5. Threat model (who's protected vs not)
docs/GOLDEN_REGRESSION_FIXTURES.md:130:## 6. Honest limits ? what a green suite does NOT prove
docs/GOLDEN_REGRESSION_FIXTURES.md:145:## 7. Deliverables & progress tracker ? **source of truth**
docs/GOLDEN_REGRESSION_FIXTURES.md:178:## 8. Open owner / counsel questions
docs/GOLDEN_REGRESSION_FIXTURES.md:194:## 9. Definition of done
docs/GOLDEN_REGRESSION_FIXTURES.md:209:## 10. References

9. user (2026-07-02T02:47:56.681Z)

```

1      <!--
2      PROJECT DOC ? tracked design plan for a deterministic "quality-neutral proof".
3      Follows docs/PROJECT_DOC_TEMPLATE.md. §7 progress tracker is the source of truth.
4      -->
5
6      # Decision-Snapshot Regression ? a deterministic quality-neutral proof
7
8      > **Status:** `DRAFT` ? **§2 decisions LOCKED (2026-06-25); ready to start §7 P1** .
**Owner:** Avi . **Created:** 2026-06-25 . **Last updated:** 2026-06-25
9      > **Lifecycle:** `DRAFT` ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
10     > **Tracking anchors:** §7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in memory `current-state` once work starts.
11     > **Relation to existing docs:** **extends** `GOLDEN_REGRESSION_FIXTURES.md` (post-
detector fixtures) and the existing characterization snapshot in
`tests/test_golden_fixtures.py`; **peer of** the M4 quality-floor track and the GPU
`scripts/nightly_reelcount.py` e2e nightly.
12     > **Honesty note:** claims tagged `[verified]` (checked vs code) / `[design]`
(proposed).
13
14     ---
15
16     ## 0. TL;DR
17     A deterministic regression that proves a change is **quality-neutral** (behavior-
preserving) by freezing and comparing the pipeline's **decision output** ? the rally segment
list **+ the stitch/concat plan** ? run off a **cached trajectory** (GPU-free). It **extends**
the existing characterization snapshot (`test_replay_matches_committed_expected`, today
segmentation-only) down to the stitch plan, runs offline in CI, and gives the **behavioral
proof** that lets a *genuine* refactor (not just `ruff format`) be confidently called quality-
neutral. **Most important caveat: never hash the encoded reel `.mp4`** ? ffmpeg/x264/NVENC/mux-
timestamps make those bytes non-deterministic, so a reel-MD5 would false-alarm on every ffmpeg
bump.
18
19     ## 1. Problem & goal
20     "Presumably quality-neutral" changes (refactors) need a cheap, deterministic proof that
they actually preserve behavior ? so they can be merged (and auto-signed-off) with confidence.
Today we have two weaker tools: **AST-identity** [verified] (used to bless `ruff format` in
#379) only proves *formatting* is unchanged ? it cannot bless a real refactor (function
extraction, rename, loop reflow) that changes the AST but should preserve behavior; and the full
**real-video F1 / reel-count e2e** needs GPU + the gitignored corpus
(`scripts/nightly_reelcount.py`) [verified] so it can't gate a PR. The decision-snapshot fills
the gap: a deterministic, offline, behavior-level proof.
21

```



```

22      **What "good" looks like:** a refactor PR runs the snapshot in CI; byte-identical
decision output ? provably quality-neutral ? eligible for auto-sign-off; a behavior-changing
edit fails with a **readable diff**.
23
24      **Read first:** `tests/test_golden_fixtures.py::test_replay_matches_committed_expected`
[verified] (the existing snapshot ? its own message: "behaviour changed vs frozen snapshot
(characterization, not correctness)**), `backend/eval/golden_fixtures.py`,
`backend/pipeline/stitcher/compiler.py` (the concat plan), audit doc **B1** (`motion`'s random
fields), `scripts/nightly_reelcount.py` (the distinct GPU e2e).
25
26      ## 2. Decisions *(D1?D11 **LOCKED** 2026-06-25)*
27
28      | # | Decision (recommendation) | Rationale | Status |
29      |---|-----|-----|-----|
30      | **D1** | **Snapshot the DECISION output, never the encoded reel.** Freeze the rally
segment list + the stitch/concat plan; never MD5 the reel `.mp4`. | Encoded mp4 bytes are non-
deterministic (ffmpeg version string, x264 threading, NVENC, mux timestamps) ? a reel-MD5 false-
alarms on every ffmpeg bump/machine. The repo's own `nightly_reelcount.py` already asserts
count+metrics, **not** bytes [verified]. | **LOCKED** (2026-06-25) |
31      | **D2** | **Input = a cached trajectory (detector output), not raw video.** Run from a
pinned trajectory CSV (the `fx_r0X` fixtures). | The WASB detector (video?trajectory) is the
GPU/WSL + CV-float-non-deterministic upstream; caching its output makes the proxy deterministic
+ CI-runnable [verified ? the fixtures are post-detector]. | **LOCKED** (2026-06-25) |
32      | **D3** | **Coverage = segmentation decision + stitch/concat plan; STOP before the
encode.** Extend the existing segmentation snapshot down to the concat manifest (clips, cut
points, order, padding). | Segmentation is already snapshotted [verified]; the stitch plan is
the deterministic missing layer. The encode is non-deterministic (D1). | **LOCKED** (2026-06-25)
|
33      | **D4** | **Compare = canonical-JSON snapshot with a readable diff** (an MD5 of the
canonical JSON may be a fast-path summary, but the field-level diff is the artifact). | A
refactor that trips it needs a debuggable diff, not "hash changed" ? mirrors
`test_replay_matches_committed_expected`'s mismatch report [verified]. | **LOCKED** (2026-06-25)
|
34      | **D5** | **Snapshot only deterministic segmenters/fields; exclude the random
fabricated fields.** Use a deterministic segmenter (CONTROL/fusion on trajectory) and a
`{start,end,shuttle}`-style schema that excludes `motion`'s random
`server/receiver/shot_count/avg_speed`. | `motion` fabricates random fields (audit **B1**)
[verified] ? non-snapshot-able as-is. **B1's fix (drop the fabrication) is the enabler that
would make `motion` snapshot-able.** | **LOCKED** (2026-06-25) |
35      | **D6** | **The reel mp4 gets a FUNCTIONAL smoke, not a byte hash** (exists, ~duration
within tol, N segments, sane streams). | Corollary of D1 ? verify the encode is *sane*, not
*identical*. | **LOCKED** (2026-06-25) |
36      | **D7** | **Runs in per-PR CI, offline, CONTROL-only ? a refactor gate, not a quality
floor.** | The point is catching a behavior change in a refactor *before merge*; it must be
fast, GPU-free, on the pinned CONTROL config. | **LOCKED** (2026-06-25) |
37      | **D8** | **Intentional changes re-freeze WITH a documented rationale (Launch
Report).** A `--update-snapshot` flag exists, but a mismatch on an *intended* change must be re-
frozen with the Launch-Report rationale recorded. | Without the discipline a snapshot becomes a
rubber stamp; ties to the launch-report convention. | **LOCKED** (2026-06-25) |
38      | **D9** | **Scope boundary: this is the quality-NEUTRAL proxy ? distinct from the M4
quality-FLOOR (per-slug F1), Tier-2 real-video e2e (GPU), and `nightly_reelcount.py`.** Keep
them separate + complementary. | Conflating "behavior unchanged" with "quality is good enough"
muddies both. | **LOCKED** (2026-06-25) |
39      | **D10** | **Deterministic-segmenters-only, ENFORCED by a determinism guard.** Cover
only deterministic segmenters; to ensure* no non-deterministic one slips in (? a flaky
snapshot), the harness runs each fixture **twice and asserts byte-identical decision output**
before freezing ? any segmenter whose two runs differ (hidden `random`, set-ordering, float/time
non-determinism) is **rejected from snapshot scope** and flagged. Mechanical, not a hand-
maintained allowlist. | Owner, 2026-06-25. `motion` is non-deterministic until **B1 / #385**
lands; then it auto-becomes eligible (no allowlist edit). | **LOCKED** (2026-06-25) |
40      | **D11** | **Auto-sign-off integration is STAGED: ADVISORY first ? GATING later.** Ship
the snapshot as an advisory/informational check; graduate it to a **gating** check (gates a non-
AST-identical refactor's auto-LGTM) only after it's proven stable on real refactors. | Owner,
2026-06-25. Resolves S8 Q1 ? the strategic payoff, de-risked by starting advisory. | **LOCKED**
(2026-06-25) |

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41
42     ## 4. Design / architecture
43     **Flow (all offline, deterministic):**
44     `pinned trajectory CSV ? [segmentation: CONTROL segmenter] ? segment list ?
[deterministic stitch-PLAN builder: clips + cut points + padding + order, NO ffmpeg] ? canonical
JSON {schema_version, segments[], stitch_plan{}} ? MD5 / field-diff vs frozen golden`.
45     Mismatch ? behavior changed (decide if intended; re-freeze via D8).
46
47     **Reuse map** [verified unless noted]:
48     - `tests/test_golden_fixtures.py::test_replay_matches_committed_expected` +
`backend/eval/golden_fixtures.py` ? the existing characterization snapshot at the
**segmentation** layer; extend its `expected` schema + replay to also carry the stitch plan.
**(reuse + extend)**
49     - `eval_baselines/fixtures/` (`fx_r0X` + `manifest.json`) ? the pinned post-detector
trajectories. **(reuse)**
50     - `backend/pipeline/stitcher/compiler.py` ? builds the ffmpeg concat today; **[design]**
factor out a pure `build_stitch_plan(segments, cfg) -> dict` (the *decision*) from the *encode*,
so the plan is hashable without running ffmpeg. **(new seam)**
51     - `eval_baselines/` snapshot store + the cross-OS enforcement already used by the golden
suite. **(reuse)**
52
53     ## 6. Honest limits ? what this does NOT do
54     - Does **not** prove *quality is good* ? only that *behavior is unchanged* vs the frozen
golden (that's M4 / F1-floor's job).
55     - Does **not** cover the **encode** (pixels/bytes) ? only the decision + stitch plan;
the reel gets a functional smoke only (D6).
56     - Does **not** run the **real detector** (GPU/WSL) or real videos ? it runs from cached
trajectories (Tier-2 / `nightly_reelcount.py` cover that, GPU-gated).
57     - A **green snapshot does not bless a behavior change** ? it only certifies *no* change;
an intended change must be re-frozen with a rationale (D8).
58     - Only as good as the **fixtures' coverage** ? pipeline paths not exercised by `fx_r0X`
aren't protected.
59
60     ## 7. Deliverables & progress tracker ? **source of truth**
61     Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. One small PR per row,
branched from `origin/master`.
62
63     | ID | Deliverable | Depends on | Gated? | Status | PR |
64     |----|-----|-----|-----|-----|----|
65     | P0 | This doc ? decisions in S2 locked | ? | Owner | ? | (this PR) |
66     | P1 | Canonical decision-snapshot schema + deterministic serializer (segments + stitch
plan; excludes non-det fields) | P0 | No | ? | ? |
67     | P2 | Factor `build_stitch_plan()` out of `compiler.py` (pure, ffmpeg-free) + unit test
| P0 | No | ? | ? |
68     | P3 | Extend `golden_fixtures` / `test_golden_fixtures` to capture + compare the stitch
plan **+ the D10 determinism guard (run each fixture twice ? assert byte-identical; reject non-
deterministic segmenters)** | P1, P2 | No | ? | ? |
69     | P4 | Freeze CONTROL snapshots for the existing `fx_r0X` fixtures | P3 | No | ? | ? |
70     | P5 | `--update-snapshot` flow + Launch-Report hook (D8) | P3 | No | ? | ? |
71     | P6 | Wire the snapshot as a per-PR CI characterization gate (CONTROL-only) | P4 | No |
? | ? |
72     | P7 | **Surface the snapshot on refactor PRs ? ADVISORY** (informational; does not
gate) ? per D11 | P6 | No | ? | ? |
73     | P8 | **(gated)** Graduate to GATING auto-sign-off: a snapshot-match marks a non-AST-
identical refactor quality-neutral ? LGTM-eligible (only after P7 proven stable) ? per D11 | P7
| **Owner** | ? | ? |
74
75     ## 8. Open questions ? owner
76     1. ? **RESOLVED (D11):** advisory first ? graduate to gating. *(The strategic payoff,
de-risked by staging.)*
77     2. ? **RESOLVED (D10):** deterministic-segmenters-only, **enforced by the double-run
determinism guard** ; **B1 is an enabler, not a blocker** ? and it's being fixed in **385**,
which makes `motion` auto-eligible.
78     3. **(still open) Fixture scope:** reuse the existing `fx_r0X` (which stop at
segmentation) by adding a stitch-plan `expected`, or add a dedicated fixture set?

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79      4. ? **DONE: D1?D11 LOCKED 2026-06-25.**
80
81      ## 9. Definition of done
82      - [x] D1?D11 locked by the owner (2026-06-25).
83      - [ ] A behavior-preserving refactor passes; a behavior-changing edit fails with a
**readable diff**.
84      - [ ] Runs **offline + deterministically** in per-PR CI, cross-OS (matching the golden
suite).
85      - [ ] Covers **segmentation + stitch plan**; the reel gets a functional smoke only.
86      - [ ] ``--update-snapshot`` requires a documented Launch-Report rationale.
87      - [ ] `docs/README.md` index + `docs/NEXT_STEPS.md` updated; **$7 all ?** ? flip `Status
? DONE`, archive.
88
89      ## 10. References
90      **Internal code `[verified]`:** `tests/test_golden_fixtures.py`
(`test_replay_matches_committed_expected`), `backend/eval/golden_fixtures.py`,
`eval_baselines/fixtures/`, `backend/pipeline/stitcher/compiler.py`,
`backend/pipeline/segmenters/motion.py` (B1 random). **Internal docs:**
`GOLDEN_REGRESSION_FIXTURES.md`, `CODE_CLEANUP_AUDIT_2026-06-24.md` (B1), `docs/NEXT_STEPS.md`
(M4 quality-floor), `PROJECT_DOC_TEMPLATE.md`; `scripts/nightly_reelcount.py` (distinct GPU
e2e). **Related:** the AST-identity sign-off used on #379; the Tier-1 nightly (#381).
91
92      ---
93
94      ### Changelog
95      - `2026-06-25` ? Initial draft from the "MD5-proxy-for-e2e validation" discussion;
decisions **D1?D9 proposed for locking**.

```

10. user (2026-07-02T02:47:57.669Z)

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145      ## 7. Deliverables & progress tracker ? **source of truth**
146
147      Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. One **small PR per row**,
branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.
148
149      | ID | Deliverable | Depends on | Counsel-gated? | Status | PR |
150      |----|-----|-----|-----|-----|----|
151      | **P0** | **PII leak hygiene (partial ? paths + scan).** Scrubbed personal `C:\Users\`?
paths from code docstrings (`eval_partial_labels.py`, `gemini_refine.py`, `README.md`) + added
`tools/leak_scan.py` (32-hex MD5 + personal-path **structural** guard, optional gitignored
`.leakscan_private` name-list) + a CI `leak-scan` job + tests. ? The **functional video-name
rename** (`eval_baselines/*.json` keys / `rally_seg_eval.MULTICOURT_CLIPS` / quality-log names)
is **DEFERRED** per the owner's safe-scope decision (2026-06-16) ? it's coupled to the
gitignored `output/human_lovo_manifest.json` + data files + the training floor and collides with
the active quality track; run the private-list scan + a coordinated rename before open-sourcing.
Git-history rewrite (PR #77) also deferred. | ? | No | ? | _this PR_ |
152      | **M1** | **Fixture framework + synthetic Tier-0 + shared replay helper** (also
delivered M2-core + M3 ? see below). `backend/eval/golden_fixtures.py` (`replay_fixture`, schema
dispatch, `Path(__file__)`-relative) + 3 **tool-generated** synthetic fixtures + manifest +
README + `.gitattributes` LF pin + `tests/test_golden_fixtures.py`. CONTROL-only (no numpy
execution on the replay path); cross-OS verified. | ? | No (synthetic) | ? |
[#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) |
153      | **M2** | **Freeze/refresh extractor.** Core (`freeze_synthetic` / `refresh_snapshot` /
`--check` CLI + idempotence test) shipped in M1 (#189); the `--license`/`--minors-
free`/`--owned` **refusal gate** shipped with P1 (this PR). | M1 | No | ? |
[#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) · _this PR_ |
154      | **M3** | **Characterization gate** ? `tests/test_golden_fixtures.py`: parse-compared
(exact-int / approx-float 1e-6), **CONTROL-only**, positive no-person/audio assertion,
perturbation + idempotence. **Shipped in M1 (#189).** (Windowing/label fingerprint guard + row-
count cap ? fold into M4.) | M1 | No | ? | [#189](https://github.com/avidullu/badminton-
highlight-indexer/pull/189) |
155      | **MX** | **Cross-OS determinism guard (CI).** Lightweight `golden-crossos` CI job
(ubuntu/windows/macOS) running `--check` + the fixtures test ? locks the cross-OS guarantee
(CONTROL replay is numpy-only, no GPU stack; a *separate fast job*, not full-suite×3). | M1 | No
| ? | [#191](https://github.com/avidullu/badminton-highlight-indexer/pull/191) |

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156 | **M4** | **Quality-floor gate + first `regression_baseline.json`.**
`tests/test_golden_quality_floor.py`; create `eval_baselines/regression_baseline.json` with
**windowing fingerprint**;; metrics-vs-base paired-delta; corpus-level (not per-slug) LOVO for
runtime. | M1, M2 | No | ? | ? |
157 | **P1** | **De-attribution + minimization** ? `golden_fixtures.freeze_real_fixture`()
(extends the M1 module, not a separate script): **opaque random `fx_hex` surrogate slug**,
metadata strip, metric-invariant **time-origin reset**, strict GT loader, **positive no-
person/audio/MD5/source-path assertion**, manifest non-clobber; `--freeze-real` CLI with the
**refusal gate** (exit 2 unless owned + minors-free + license). Tooling-only, 12 synthetic
tests; **no real data committed**. | M1 | No (tooling) | ? | _this PR_ |
158 | **P2** | **Owner-accepted (researched-risk), 2026-06-16.** Owner accepted the
researched IP/privacy analysis (§3) to proceed **under its conditions** (owned/Fork-A source .
minors excluded . biometric/person streams excluded . de-attributed). Owner's informed risk
acceptance ? **NOT a retained-counsel opinion**;; optional future counsel confirmation per §8. |
P1 | Owner-accepted | ? | ? |
159 | **P3** | **Tier-0 REAL subset ? SHIPPED.** Committed **6 curated, trimmed (?12k-frame
? few-min), de-attributed** real fixtures (`fx_r01`?`fx_r06`: single-court / multi-court×2 /
doubles×2 / continuously-active blob; both 29.97 & 59.94 fps) via `freeze_real_fixture --max-
frames` (opaque slug, time-origin reset, scenario-only labels, **NO** source name/path/video_id
? grep-verified + leak-scan clean). ~1.9 MB total; the characterization + cross-OS gates now run
on real data too. Owner cleared ALL owned golden videos (owned + minors-free) + re-confirmed
option (a) (2026-06-22). The surrogate?source map stays **off-git** (owner-held). | P1, P2 |
conditions §3 ? | ? | _this PR_ |
160 | **O1** | *(optional)* **Protobuf + deterministic serialization.** `rally.fixture.v1`
proto3, codegen + `protoc`/`git diff --exit-code` guard, canonical writer, `.gitattributes`
binary. Only if cross-OS byte-stable snapshots are wanted. | M1 | No | ? | ? |
161 | **O2** | *(optional)* **Tier-1 key-gated path.** `scripts/fetch_golden.py` (OneDrive
token, default) and/or SOPS+age; `_tier1_present()`; `RALLY_GOLDEN_DIR` wired; Tier-1 CI job
gated on secret, never on fork runners. | M1 | No | ? | ? |
162 | **DOC** | **Discoverability cross-links** ? added the `eval_baselines/fixtures/` row
to `CODE_MAP §0` + a `golden_fixtures.py` entry to `CODE_MAP §2.3`; cross-linked
`GOLDEN_DATA_SHARING.md` (committed subset ? private OneDrive corpus). README indexed in #186;
memory `current-state` kept live. (EVALUATION/QUALITY_ITERATIONS cross-links **deferred** ? that
doc surface is hot in the concurrent quality track; avoid collision.) | M1 | No | ? | _this PR_
|
163
164 **Must-fix-before-any-public-commit (red-team), folded into the rows above:** ? numeric-
tolerance not byte-equality + multi-OS matrix + CONTROL-only (M3/D5/MX ?); ? scrub names in
**all** files + history decision (P0); ? leak-scan a **required** check, hooks are convenience
only (P0); ? person-optional loader + **positive** no-person assertion, don't trust the silent
`0.0` default (P1 ?); ? random surrogate id not HMAC-over-roster (P1 ?); ? quantize/drop
fps/frame_width (P1 ? fps/fw retained as scale constants; quantize is a P3 option); ? document
Tier-0 omits the fusion path (DOC); ? counsel + Fork-A + minors/biometric exclusion as hard
preconditions (P2 owner-accepted; enforced in code by P1's refusal gate ?).
165
166 ### Progress log (quantitative ? coverage must hold/improve, CI incl. the 3-OS cross-OS
job green)
167 | PR | Deliverable(s) | Suite | Coverage | Notes |
168 |----|-----|-----|-----|-----|
169 | [#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) | M1 +
M2-core + M3 | 746 pass / 7 skip | 80.41% | synthetic Tier-0 + CONTROL replay +
characterization; cross-OS proven (CI Linux replayed windows-frozen fixtures) |
170 | [#191](https://github.com/avidullu/badminton-highlight-indexer/pull/191) | MX +
tracker/legal reconcile | full suite green | 80.41% | `golden-crossos` CI job green on
ubuntu+windows+macos ? cross-OS **enforced**;; legal ? owner-accepted |
171 | _this PR_ | P1 + M2 refusal gate | 759 pass / 7 skip | **80.49%** (+0.08) |
`freeze_real_fixture` + refusal gate; **+13 tests**;; tooling-only, no real data committed |
172 | _this PR (audit-hardening)_ | adversarial-audit fixes #1?#17 | 784 pass / 7 skip |
**80.71%** (+0.22) | strict-slug `label` + dropped `source_note` (#1); 6dp-rounded reset within
`_FLOAT_TOL` (#2); temp-dir-then-promote + negative-label refusal (#3); exact-membership source-
path scan (#4); key-set-equality in `compare_expected` (#5); `--reason` mandatory (#6);
independent schema guard (#7); `max_win` recorded-not-applied note (#8); numpy-execution wording
(#9); +failure-semantics & by-kind tests (#10/#11); `test_golden_real_fixtures.py` added to
`golden-crossos` CI (#12); doc reconciled to shipped reality incl. per-frame-CSV honesty + risk
re-confirm (#13?#17). Synthetic `--check` byte-identical (unchanged). |

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173 | _this PR (P0 safe scope)_ | personal-path scrub + leak-scan guard | 1365 pass / 7 skip
| 85.27% | scrubbed `C:\Users\?` paths from 3 code docstrings (`eval_partial_labels.py` /
`gemini_refine.py` / `README.md`); `tools/leak_scan.py` (32-hex MD5 + personal-path structural
guard + gitignored `.leakscan_private` name-list) + a CI `leak-scan` job + 10 tests. Functional
video-name rename DEFERRED (owner safe-scope). |
174 | _this PR (P3 real fixtures)_ | 6 curated trimmed de-attributed real Tier-0 fixtures +
`--max-frames` trim | 1390 pass / 7 skip | 85.27% | `fx_r01`?`fx_r06` (~1.9 MB; single/multi-
court/doubles/blob, both fps) frozen via `freeze_real_fixture --max-frames=12000`; de-
attribution grep-verified + leak-scan clean; +trim test; owner re-confirmed option (a). F1
0.0?0.59 = real CONTROL-path (behaviour-locking, **NOT** field quality). |
175
176 ---
177
178 ## 8. Open owner / counsel questions
179
180 **Counsel questions ? documented basis (the owner accepted the researched risk on
2026-06-16; see P2). These remain the record, and a retained-counsel confirmation is
optional/recommended before scaling beyond a small owned, minors-free, de-attributed subset:**
181 1. Does the EU/UK **sui-generis database right** subsist in our self-derived
shuttle/rally streams on the *Sportradar* "obtained vs created" reasoning, and does
redistributing a de-attributed Tier-0 subset cause *CV-Online* harm?
182 2. Post *Thomson Reuters v. Ross* + USCO-2025, is **US fair use for the extraction
step** defensible if the dataset could compete in a "data/analytics" market ? does publishing
only numbers (never footage) matter?
183 3. India: are anti-scraping ToS enforceable; **IT Act s.43/s.66** exposure for any 3rd-
party ingestion; does pending *ANI v. OpenAI* (~Apr 2026) move it? Confirm **DPDP s.9/Rule 10**
and whether per-player tracking is prohibited s.9(3) monitoring of a minor.
184 4. Sign-off on the **final committed Tier-0 subset + provenance log**; is non-
attributability-via-private-source a defensible posture to assert given its
conditional/revocable nature?
185
186 **Owner (strategic):**
187 5. Publishing the Fork-A subset **forfeits the trade-secret moat** (Indian breach-of-
confidence framing) ? weighed against community/benchmark/paper aims. Your call.
188 6. **Forward-looking commitment:** do not publish a highlight reel of the *same* source
whose fixtures are committed (without re-clearing), and accept a periodic re-assessment as the
public-video environment grows.
189 7. **Scope of first public commit:** synthetic-only (zero provenance risk, recommended)
vs a counsel-cleared real subset.
190 8. **Tier-1 mechanism:** out-of-repo+token (recommended) vs SOPS+age in-repo. And: is
protobuf (01) wanted, or is canonical-JSON enough?
191
192 ---
193
194 ## 9. Definition of done
195
196 The project is **DONE** (flip `Status ? DONE`, archive) when:
197 - [ ] **P0, M1?M4, P1, DOC** are ? and merged.
198 - [ ] A clean `git clone` on **win + ubuntu + mac** runs `python run_all_tests.py` and
the **Tier-0 gates pass green with no key / no video / no GPU / no DB**.
199 - [ ] The **CI leak-scan is a required status check** and passes (no MD5 / real-name /
non-surrogate / plaintext-SOPS in the tree).
200 - [ ] Characterization + quality-floor gates are **parse-compared, CONTROL-only,
windowing-fingerprinted**, and demonstrably catch a seeded behaviour change.
201 - [ ] `docs/CODE_MAP.md §0` + TEST MAP updated; this doc cross-linked; `NEXT_STEPS.md` +
memory `current-state` point here.
202 - [ ] **Either** the public commit is synthetic-only, **or** (**P2 owner-accepted ? +
P3**) for a real-data commit, under the §3 conditions enforced **in code** by P1's refusal gate.
203 - [ ] §6 honest-limits caveat rides in the test pass/fail messages, not only the docs.
204
205 *Optional 01/02 may remain ? without blocking DONE (they're earned, not assumed).*
206
207 ---
208
209 ## 10. References

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210
211     **Internal code anchors (verified):** `backend/eval/{regression.py,fusion_golden.py,fusi
on_compare.py,ablation.py,experiment.py,metrics.py>windowing.py,distill_local.py,gt_loader.py,eva
al_stats.py}` · `backend/pipeline/detectors/{base.py,tracknet_runner.py,rally_gate.py}` ·
`backend/utils/hashing.py::compute_video_id` · `eval_baselines/{heuristic_lovo_n6.json,gemini_wr
apper_2026-06-10.json,experiment_leaderboard.json}` ·
`tests/test_ablation.py::test_litmus_reproduces_gate_a` · `.github/workflows/ci.yml`.
212     **Internal docs:** `GOLDEN_DATA_SHARING.md` (#175), `EVALUATION.md`,
`EXPERIMENT_HARNESS.md`, `QUALITY_ITERATIONS.md`, `MULTI_SIGNAL_FUSION_PLAN.md`, `CODE_MAP.md
§0`.
213     **Legal authorities:** see §3 (full URLs in the research workflow artifact). *This
section is researched general information, not legal advice.*
214
215     ---
216
217     ### Changelog
218     - **2026-06-16** ? Initial draft synthesized from two multi-agent workflows (mechanics
design + 2 critiques; legal memo IN/US/EU + SEALED-FIXTURES architecture + red-team). Verified
the live name-leak and the BLAS-determinism cause against code. Status: DRAFT, owner-gated.
219

```

11. assistant (2026-07-02T02:48:21.210Z)

Now let me verify these doc claims against current code. Starting with PR #390 and the main verification targets.

12. user (2026-07-02T02:48:24.382Z)

```

{"base":"master","createdAt":"2026-06-26T14:28:06Z","head":"refactor/p10-split-main","mergeState
Status":"DIRTY","mergeable":"CONFLICTING","nfiles":3,"state":"OPEN","title":"refactor(D1/P10):
extract orchestration.py from main.py (god-module split, step
1)","updatedAt":"2026-06-26T14:37:18Z"}

```

13. user (2026-07-02T02:48:25.395Z)

```

2198 backend/main.py
ls: cannot access 'tests/test_golden_quality_floor.py': No such file or directory
ls: cannot access 'eval_baselines/regression_baseline.json': No such file or directory

```

14. user (2026-07-02T02:48:37.469Z)

```

backend/pipeline/detectors/person_detector.py
backend/pipeline/segmenters/yolo_hybrid.py
---
requirements.txt:18:supervision>=0.21.0
requirements.txt:19:# D13/P12: ByteTrack moved out of `supervision` (removed in 0.30.0) to the
standalone
requirements.txt:20:# `trackers` package. The PyPI wheel is BROKEN (ships metadata only, no
source ? see
requirements.txt:21:# https://github.com/roboflow/trackers/issues/474), so install from git
until fixed:
requirements.txt:22:#   pip install "trackers @
git+https://github.com/roboflow/trackers.git@2.5.0"
requirements.txt:23:# (Apache-2.0; depends on supervision >=0.26.1, so it composes cleanly.)
requirements.txt:24:# NOTE: roboflow/trackers uses bare version tags (2.5.0), NOT v-prefixed.
requirements.txt:29:trackers @ git+https://github.com/roboflow/trackers.git@2.5.0
pyproject.toml:36:     "supervision>=0.21.0",
pyproject.toml:38:     # maximum_filter1d); previously present only TRANSITIVELY via supervision,
so a supervision
pyproject.toml:41:     # D13/P12: ByteTrack moved to the standalone `trackers` package
(Apache-2.0). Its
---
backend/pipeline/detectors/person_detector.py:12:`trackers.ByteTrackTracker` with its
constructor params pinned to `sv.ByteTrack`s

```

backend/pipeline/detectors/person_detector.py:109:	to the standalone ``trackers`` package
(``ByteTrackTracker``). Imported lazily so	
backend/pipeline/detectors/person_detector.py:112:	The constructor params are pinned to
``sv.ByteTrack``'s behaviour so the migration	
backend/pipeline/detectors/person_detector.py:124:	from trackers import ByteTrackTracker
backend/pipeline/detectors/person_detector.py:127:	return ByteTrackTracker()

15. user (2026-07-02T02:48:38.464Z)

Exit code 2

```
64: def _parse_time(val: Union[int, float, str]) -> float:
113:     s = VideoCompiler._parse_time(raw_s)
114:     e = VideoCompiler._parse_time(raw_e)
45
ls: cannot access 'backend/annotations.py': No such file or directory
backend/eval/annotations.py
```

16. user (2026-07-02T02:48:48.507Z)

```
on-by-default bundled-font watermark (and absolute logo paths) on Windows.""
return path.replace("\\", "/").replace("'", r"\'").replace(":", r"\:")
```

```
@staticmethod
def _parse_time(val: Union[int, float, str]) -> float:
    """Normalizes a timestamp value to float seconds.
    Handles: float/int (pass-through), plain numeric strings ("12.4"),
    and colon-separated strings ("MM:SS", "HH:MM:SS", "H:MM:SS.ms").
    This mirrors the parse_time logic used in gemini.py to ensure
    the stitcher can always consume whatever format the indexer produced."""
    if isinstance(val, (int, float)):
        return float(val)
    if isinstance(val, str):
        val = val.strip()
        if ":" in val:
            # Handle MM:SS, HH:MM:SS, etc.
            parts = val.split(":")
            parts.reverse()
            total = 0.0
            for i, p in enumerate(parts):
                try:
                    total += float(p) * (60**i)
                except ValueError:
                    pass
            return total
        try:
            return float(val)
        except ValueError:
            logger.warning(
                f"Could not parse timestamp value '{val}' as a number. Defaulting to 0.0."
            )
            return 0.0
    logger.warning(
        f"Unexpected timestamp type {type(val).__name__} for value '{val}'. Defaulting to 0.0."
    )
    return 0.0
```

```
@staticmethod
def _merge_overlapping(
    segments: List[Tuple[float, float]],
) -> List[Tuple[float, float]]:
    """Normalize, sort, and merge overlapping/duplicate (start, end) ranges.
```

A highlight reel must never replay the same footage. The same rally can land in the segment list more than once ? e.g. two detector runs accumulated for one

```

video (add_video is an UPSERT, so re-running without a prune leaves both sets), or
chunk-overlap duplicates near boundaries. Stitching those verbatim plays the rally
twice. Merging any range that overlaps or abuts the previous one collapses true
duplicates/overlaps while preserving genuinely distinct (gapped) rallies.
Degenerate rows (end <= start) are dropped. Timestamps may be float/int/str.
"""
parsed = []
for raw_s, raw_e in segments:
    s = VideoCompiler._parse_time(raw_s)
    e = VideoCompiler._parse_time(raw_e)
    if e > s:
        parsed.append((s, e))
parsed.sort()
merged: List[Tuple[float, float]] = []
for s, e in parsed:
    if merged and s <= merged[-1][1]: # overlaps/abuts previous -> extend it

```

17. user (2026-07-02T02:48:49.162Z)

```

__pycache__
annotations
api
assets
audio
billing.py
compute
config
detectors
eval
features
flywheel.py
infrastructure
main.py
pipeline
providers
reporting
results.py
sports
stitcher
storage
tools
ui
utils
worker
---
backend/annotations
backend/eval/annotations.py
backend/eval/__pycache__/annotations.cpython-313.pyc
backend/utils/annotation_proxy.py

```

18. user (2026-07-02T02:49:09.358Z)

```

15:         return "Analyzes Presentation Time Stamps (PTS) of keyframes to check for timeline
gaps, jumps, or editing splices."
66:         # Analyze gaps
67:         max_gap = 0.0
69:         gaps = []
72:         gap = timestamps[i] - timestamps[i - 1]
73:         gaps.append(gap)
74:         if gap < 0:
76:         if gap > max_gap:
77:             max_gap = gap
80:         # of 1.0 to 4.0 seconds. A gap larger than the configured threshold
---B7---
46:     driver's legacy lenient semantics (skip bad rows), but skipped rows are now logged.

```



```

---B8---
56:     regressed: bool
139:     If no baseline exists yet, `regressed` is False (nothing to compare to) and the
150:     regressed = False
163:         regressed = True
168:         regressed = True
173:         regressed = True
178:         regressed = True
183:         regressed = True
200:         regressed=regressed,
243:     verdict = "REGRESSED" if r.regressed else "ok"

```

19. user (2026-07-02T02:49:10.790Z)

backend/compute:

```

__init__.py
__pycache__
base.py
cloud_run.py

```

backend/detectors:

```

__pycache__

```

backend/stitcher:

```

__pycache__

```

backend/worker:

```

__init__.py
__main__.py
__pycache__

```

```

---
```

```

22
e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
81a78bf api: /api/videos returns video_id + a friendly match_name (fix disappearing name) (#412)
8b16cf4 docs: reflect today's storage endpoints + the corpus/user-video boundary (#411)
aa915a7 deploy/gcp: centralize project + bucket names in one sourced buckets.env (#410)
52d97b5 P4: POST /api/assets ? store a video into the user's chosen medium (#409)
7ea6ea9 P3: /api/capabilities reports usable storage backends (#408)
f28af73 P2: engine free-space guard (HTTP 507 + WSL frame-stage BLOCK) (#407)
759a67f feat(api): frame-preserving annotation proxy for /api/source (D4) (#406)
5a5c4fa feat(api): P8a ? GET /api/source + POST /api/annotations (annotate-in-UI backend) (#405)
4a7bfbe docs: record the launch NO-GO decision (owner, 2026-06-30) (#404)
6e973af docs: fix stale rally-cluster claims (no archival ? cluster still gated) (#403)
af1ccbe chore(#399): launch-PR process hardening ? link-check CI, PR template, doc cross-links
(#402)
c8a493d fix(#398): served Gate-A litmus passes a typed IndexingConfig (not a dict) (#401)
959c630 fix(#397): write per-video report to the resolved local path, not the raw URI (#400)
39dff1d feat(config): promote inactivity_temporal_density to default-ON (P2b) (#396)
0c9c693 feat(eval): expose inactivity_temporal_density in the rally_seg_eval A/B CLI (P2a)
(#395)
cc60ebe Fix/rally detection edge cases (#394)
69c6266 docs(nightly): add person-cue Stage-1 stats lane to the engine-regression plan (design)
(#389)
43578a7 chore(gcp): teardown.sh accepts RALLY_VM_NAME (align with create_gpu_vm.sh) (#392)
ffb1f53 ci: harden the 'System libs' apt step against the runner's Microsoft-repo 403 flake
(#391)
b6214e5 chore: ruff format the whole repo (259 files) (#387)
3624a8d feat(D13/P12): migrate ByteTrack to standalone trackers package (#386)
0b4f492 fix(B1/P9): remove fabricated data from motion segmenter (#385)
1283e18 docs(project): Decision-Snapshot Regression plan ? a deterministic quality-neutral proof
(#383)
9972057 ci: add Tier-1 nightly quality-regression workflow (#381)
3a34c42 ci: add per-job timeout-minutes to every ci.yml job (hardening) (#382)
cd5b3d2 feat(01/P11): Gemini remote-file GC ? startup orphan sweep (#380)
86705c1 chore: ruff format gemini_refine.py + trajectory_hybrid.py (#379)

```

b72ac85 fix(B2/P17): surface Gemini provider failures in all callers (#378)
71a8241 chore(repo): fix corrupted .gitignore (UTF-16 tail) + ignore artifacts/, track product overview (#377)

20. user (2026-07-02T02:49:49.246Z)

```
25:# 01/P11: every upload is tagged with this display_name prefix so the startup orphan
28:# are ever deleted by cleanup_orphaned_files().
59:     def cleanup_orphaned_files(self, grace_sec: int = 3600) -> int:
60:         """Sweep and delete video files orphaned on the Gemini File API by a previous
107:            "[Gemini GC] deleted %d orphaned video file(s) older than %ds.",
---
---
52:MULTICOURT_CLIPS = frozenset(
59:     ``court_type`` field when present, else falls back to membership in ``MULTICOURT_CLIPS``
64:     return "multicourt" if video.get("name") in MULTICOURT_CLIPS else "single"
52:MULTICOURT_CLIPS = frozenset(
53-     {"mahadevpura_2", "GX010128", "Badminton_BXH_2", "Boxhill_Doubles"}
54-)
55-
56-
57-def court_type_for(video: Dict[str, Any]) -> str:
58-     """``multicourt`` or ``single`` for a golden video. Honors an explicit manifest
```

21. user (2026-07-02T02:49:50.913Z)

```
{"checks":[{"conclusion":"SUCCESS","name":"Ruff lint"}, {"conclusion":"SUCCESS","name":"Leak scan
(PII / attribution guard)"}, {"conclusion":"SUCCESS","name":"Mypy (lenient
baseline)"}], "files":["backend/main.py", "backend/orchestration.py", "mypy.ini"], "ncomments":0}
```

22. user (2026-07-02T02:50:06.878Z)

```
14:# backend.api.job_worker lifted from quarantine (now takes explicit config/db params).
15:[mypy-backend.main]
19:# backend.main, which are typed `config: dict` but receive the AppConfig migration shim
20:# at runtime (the .get() compat layer). Shares backend.main's config-shim quarantine
---
e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
81a78bf api: /api/videos returns video_id + a friendly match_name (fix disappearing name) (#412)
52d97b5 P4: POST /api/assets ? store a video into the user's chosen medium (#409)
7ea6ea9 P3: /api/capabilities reports usable storage backends (#408)
f28af73 P2: engine free-space guard (HTTP 507 + WSL frame-stage BLOCK) (#407)
759a67f feat(api): frame-preserving annotation proxy for /api/source (D4) (#406)
5a5c4fa feat(api): P8a ? GET /api/source + POST /api/annotations (annotate-in-UI backend) (#405)
959c630 fix(#397): write per-video report to the resolved local path, not the raw URI (#400)
b6214e5 chore: ruff format the whole repo (259 files) (#387)
cd5b3d2 feat(01/P11): Gemini remote-file GC ? startup orphan sweep (#380)
---
18
```

23. user (2026-07-02T02:50:32.315Z)

```
__init__.py
__pycache__
automated_provider.py
base.py
file_provider.py
"""Pluggable sources of ground-truth rally annotations.

`AnnotationProvider`s yield `(start, end)` intervals ? the same shape the eval
harness scores ? so golden-set labels can come from a file, a human vendor, or an
automated oracle behind one interface. See `docs/GOLDEN_SET_VENDORING.md`.
"""
```

```

from . import (
    automated_provider, # noqa: F401 (side-effect: registers AutomatedProvider)
    file_provider, # noqa: F401 (side-effect: registers FileProvider)
)
from .base import AnnotationProvider, annotation_registry

__all__ = ["annotation_registry", "AnnotationProvider"]
---
"""Load hand-annotated ground-truth rally timestamps from CSV.

Annotation file format (one file per video):

    start,end
    00:01:12,00:01:39
    102.5,131.0

- A header row `start,end` is optional (auto-detected and skipped).
- Each timestamp is either decimal seconds (`102.5`) or colon-separated
  `MM:SS` / `HH:MM:SS` (`00:01:12`). Both forms may be mixed in one file.
- Blank lines and `#` comment lines are ignored.
"""

```

```

import csv
import logging
from typing import List, Tuple

logger = logging.getLogger(__name__)

Interval = Tuple[float, float]

```

```

def parse_timestamp(value: str) -> float:
    """Parse a timestamp string to float seconds.

```

24. user (2026-07-02T02:50:48.140Z)

```

6e973af 2026-06-30
b3c6c2b 2026-06-23
6e973af 2026-06-30
---
41:- [CODE_CLEANUP_AUDIT_2026-06-24.md](CODE_CLEANUP_AUDIT_2026-06-24.md) ? **? Tracked project
(DRAFT, owner-gated):** the next-wave cleanup project ? 33 audit findings across bugs, tech
debt, operational risks, and quick wins. Phase 1 = pure refactorings (typed-config migration,
app-factory, segmenter validator fix); Phase 2+ = logic fixes. §7 progress tracker is the source
of truth; sibling of the completed hardening sweep above.
42:- [DECISION_SNAPSHOT_REGRESSION.md](DECISION_SNAPSHOT_REGRESSION.md) ? **? Tracked project
(DRAFT, owner-gated ? decisions in §2 need locking):** a deterministic **quality-neutral proof**
? freeze + compare the pipeline's **decision output** (rally segments + stitch/concat plan) off
a cached trajectory, GPU-free, in CI. The behavioral proof that lets a *genuine* (non-AST-
identical) refactor be confidently called quality-neutral. **Never hashes the encoded reel**
(non-deterministic). Extends the existing characterization snapshot (`test_golden_fixtures.py`)
from segmentation to the stitch plan; peer of the M4 quality-floor + GPU `nightly_reelcount.py`.
§7 tracker is the source of truth.

```

25. user (2026-07-02T02:51:41.109Z)

```

['name', 'description', 'computed', 'mean_f1', 'std', 'per_video']
---
93: def create_app(config: AppConfig, db: Database) -> FastAPI:
---

`ground_truth` is a list of (start, end) intervals ? typically obtained from an
AnnotationProvider (see `regress_with_provider`). A regression is flagged when
precision OR recall falls more than `tolerance` below the snapshotted baseline.
If no baseline exists yet, `regressed` is False (nothing to compare to) and the

```

```
caller can choose to snapshot the current numbers via `save_baseline`.
"""
preds = get_predictions(db, video_id, stage, detector=detector)
s = metrics.score(preds, ground_truth, iou_threshold)
fp = gt_hash(ground_truth)

baselines = load_baselines(baseline_path)
b = baselines.get(_baseline_key(video_id, stage))

reasons: List[str] = []
regressed = False
has_baseline = b is not None
base_p = base_r = None
if b is not None:
    base_p, base_r = b["precision"], b["recall"]
    # Incommensurable comparison guard: a baseline taken at a different IoU/detector or
```

26. user (2026-07-02T02:51:48.503Z)

```
['adarsh_avi_singles', 'kushagra_singles', 'largetest_doubles', 'gbaaddy', 'testlarge_short',
'mahadevpura_singles']
18
45
```

27. user (2026-07-02T02:53:47.836Z)

Structured output provided successfully